

Single-Stage PFC LLC Converter Design

with the STMicroelectronics L6790A

Worked design

90 to 264 Vac, 47 to 63 Hz

25 V / 26.3 A = 657.5 W

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Before you start

The L6790A behaviour described here comes from a **draft** datasheet dated 30 April 2026. That draft has items marked TBD and at least one place where it contradicts itself. Check every controller constant against the released datasheet before the design goes to production.

The worked design has not been built. Only the numbers marked as measured are measured; the rest are calculated.

Some of the LLC theory figures are taken from the application notes in the reference list. Each one names its source in the caption.

Revision history

Version	Date	Change
V1.0	August 2026	First issue. Worked design: 25 V / 26.3 A, 657.5 W.
V1.1	September 2026	Background section on LLC and power factor correction added, with circuit figures. Transformer design, the closed form for $f_{sw}(\theta)$, the compensator design method, the gain-margin check and the input capacitor added. Worked design moved to the $N_p:N_s = 15:2$ transformer.
V1.2	September 2026	Renamed single-stage PFC LLC throughout. Verification margins defined. Worked values moved out of the design procedure and into the design example, where each one names the equation it comes from and shows the numbers put into it.

Conventions

- Frequencies are switching frequencies unless written f_l , which is the line frequency.
- $f_n = f_{sw}/f_r$ is the switching frequency normalised to the series resonance. Gain curves are always drawn against it, so that one curve serves every tank. Figures reproduced from other documents may write f_s for f_{sw} ; that is the same quantity.
- M is the voltage gain of the resonant network: the output voltage referred to the primary, divided by the fundamental of the square wave driving it. **$M = 1$ at the series resonance**, whatever the load.
- $Z_0 = \sqrt{L_r/C_r}$ is the characteristic impedance of the tank, and $Q = Z_0/R_{ac}$ is its quality factor — how heavily the tank is loaded. $Q = 0$ is no load; larger Q is more power and a flatter, lower gain curve.
- $V_{o,eff} = V_{out} + N_{rect} V_f$ is the output voltage the primary actually sees, with the rectifier drop added. N_{rect} is the number of rectifier drops in the conduction path: 1 for a centre tap, 2 for a full bridge.
- n is the equivalent-model turns ratio used in the gain equation; n_T is the physical turns ratio of a transformer whose leakage is integrated into L_r .
- $\lambda = L_r/L_m$, and $m = 1 + 1/\lambda$ is the ratio of total primary to resonant inductance used in older literature.
- θ is the line phase angle, zero at the mains zero crossing.
- “Equivalent input” is what the resonant tank sees after topology morphing, which is not the mains voltage.

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1 Introduction

A conventional universal-input LLC power supply is two stages: a boost power-factor corrector that holds a 400 V bus, and an LLC converter that steps that bus down. The capacitor between them does two jobs at once. It buffers the energy a unity-power-factor input cannot deliver smoothly, and it presents the LLC with an almost constant input, so the resonant tank only has to cover a narrow range.

A **single-stage PFC LLC** deletes both the boost stage and that capacitor. The rectified mains feeds the resonant tank directly. What results is a converter whose input is a 100/120 Hz half sine, whose gain is no longer a control variable, and whose output capacitor must absorb everything the bus capacitor used to. It is a genuinely different converter rather than an LLC with a stage removed, and most of the design habits carried over from the two-stage case are wrong here.

What follows is how such a converter works, and how to design one with the STMicroelectronics **L6790A** controller. One worked design runs through it from specification to component values: **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, a low-voltage, high-current dc rail.

Scope. Covered: the power stage, the resonant tank, the output bank, the controller network and the voltage loop.
Not covered: EMI filtering, magnetics construction, layout and safety approval.

2 The two converters this one is made of

Everything that follows is easier if the two halves are separate in your head first. This section is ordinary LLC and ordinary power factor correction; a reader who has built both can skip to Section 3.

2.1 What an LLC converter is

An LLC converter is a square-wave generator driving a resonant network that feeds a transformer and a rectifier. The switches do nothing but chop the input into a square wave of adjustable frequency — there is no duty cycle to set and no inductor current to program.

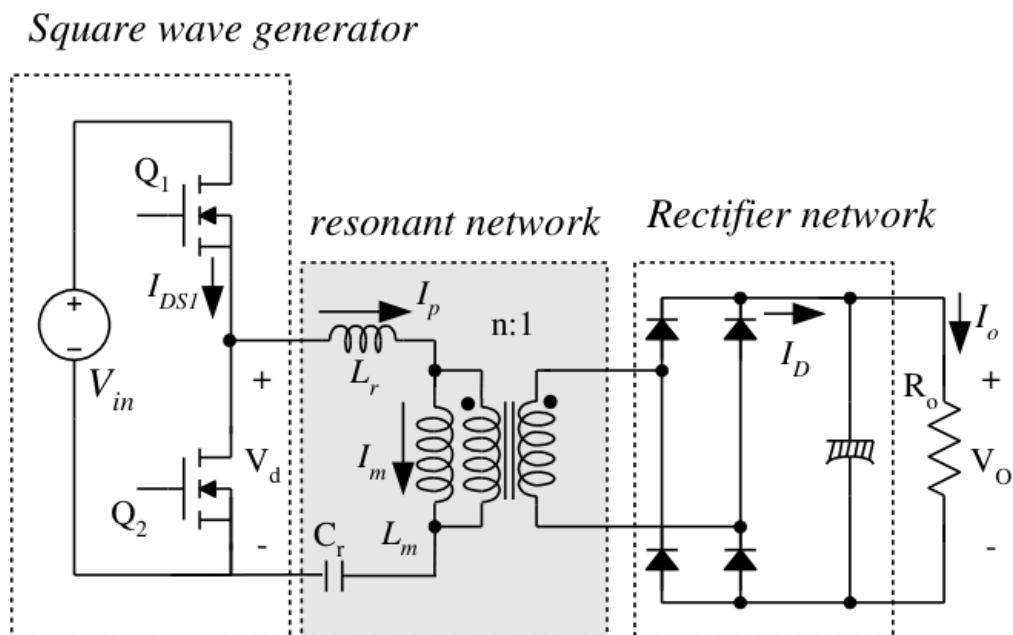


Figure 1: An LLC half bridge: a square-wave generator, the resonant network, and a rectifier. (ON Semiconductor AN-4151.)

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The three tank elements are named in the order they appear in “LLC”: the series inductance L_r , the magnetising inductance L_m of the transformer, and the series capacitance C_r . The middle element is not an added part — it is the transformer’s own magnetising inductance, which is why an LLC has one fewer component than the name suggests.

- **The square wave sets the frequency, and nothing else.** Output voltage is controlled by moving f_{sw} , because the impedance of the tank — and therefore its voltage division — depends on frequency.
- **The tank current is very nearly sinusoidal.** That is the point: the switches turn on and off while the current is small or while it is flowing through the body diode, so switching loss largely disappears.
- **The transformer does the voltage step and the isolation**, and its leakage inductance can be used as L_r rather than fought. That is the usual construction at these powers.

2.2 Why an LLC, and not something simpler

An LLC is more work than a forward or a flyback and it does not regulate by duty cycle, which is the reflex every designer has. What it buys is worth the trouble at a few hundred watts and above:

- **Soft switching over the whole load range, on both sides.** The primary switches turn on at zero volts and the secondary rectifiers, below resonance, turn off at zero current. A phase-shifted full bridge loses ZVS at light load; an LLC does not, because the magnetising current that does the work is there whatever the load.
- **Frequency can be pushed up.** Switching loss is what caps the frequency of a hard-switched converter, and with it the size of the magnetics and the filter. Take that loss away and hundreds of kHz becomes reasonable.
- **The transformer leakage is used, not fought.** In a forward converter leakage is a problem to be snubbed. Here it can be L_r , so the resonant inductor costs nothing and the snubber disappears with it.
- **No output inductor.** The rectifier feeds the output capacitor directly, which removes a wound component and its loss from the high-current side — the side where it hurts most.
- **Quiet.** Currents are sinusoidal rather than trapezoidal and the switching transitions are slow and damped, so the conducted and radiated spectrum is far smaller than a hard-switched converter of the same power.

The costs are equally real and worth stating in the same breath. Regulation is by frequency, so the operating frequency moves with line and load and the magnetics have to work over that whole range. Circulating magnetising current flows whether or not the load takes power, so light-load efficiency is not the strong point. And the design does not fall out of a duty-cycle expression — it comes from the gain curve, which is what the rest of this section is about.

2.3 Why resonance buys anything

In a hard-switched converter a switch turns on with the full input voltage across it, so the energy stored in its output capacitance is dumped into the channel every cycle. That loss rises with frequency, which is what caps the frequency and therefore the size of the magnetics.

An LLC run **above its lower resonance** presents an inductive load to the bridge. The tank current then lags the drive, and in the dead time between the two switches that lagging current discharges the mid-point capacitance on its own. The switch turns on at zero volts. **That is zero-voltage switching, and it is the entire reason for the topology.** Section 2.9 shows the mechanism and what it demands of the design.

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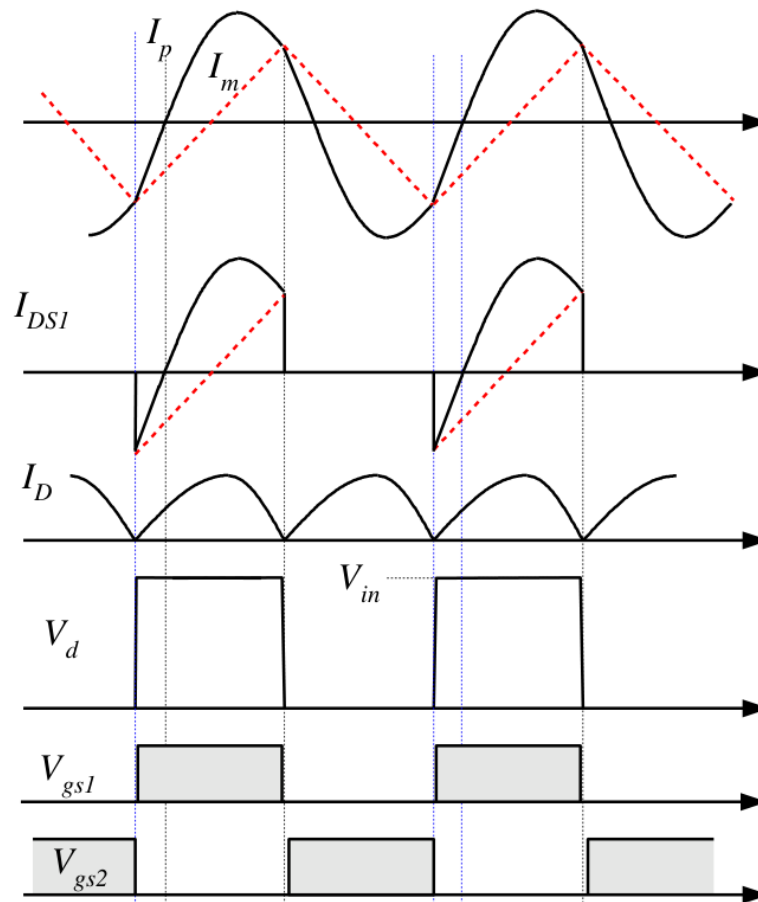


Figure 2: Typical waveforms. I_p is the tank current and I_m the magnetising current; the switch turns on while the difference between them is still flowing, which is what discharges the mid-point. (ON Semiconductor AN-4151.)

The two frequencies that bound this are the subject of Section 2.5, and the operating regions they divide are in Section 2.6.

The condition is *inductive operation*, not a particular frequency. Fall low enough and the tank turns capacitive: the current leads, the dead time works against the transition instead of for it, and the bridge hard switches into a low impedance. It is the one failure mode that destroys parts rather than just heating them. **Where that happens is not a fixed frequency** — it is the peak of the gain curve, which moves with load, and Section 2.8 is about exactly that.

2.4 How the circuit becomes $M(f_n, Q)$

Every gain curve in every LLC note comes from three reductions applied in order. They are worth doing once, because each one discards something and it is useful to know what.

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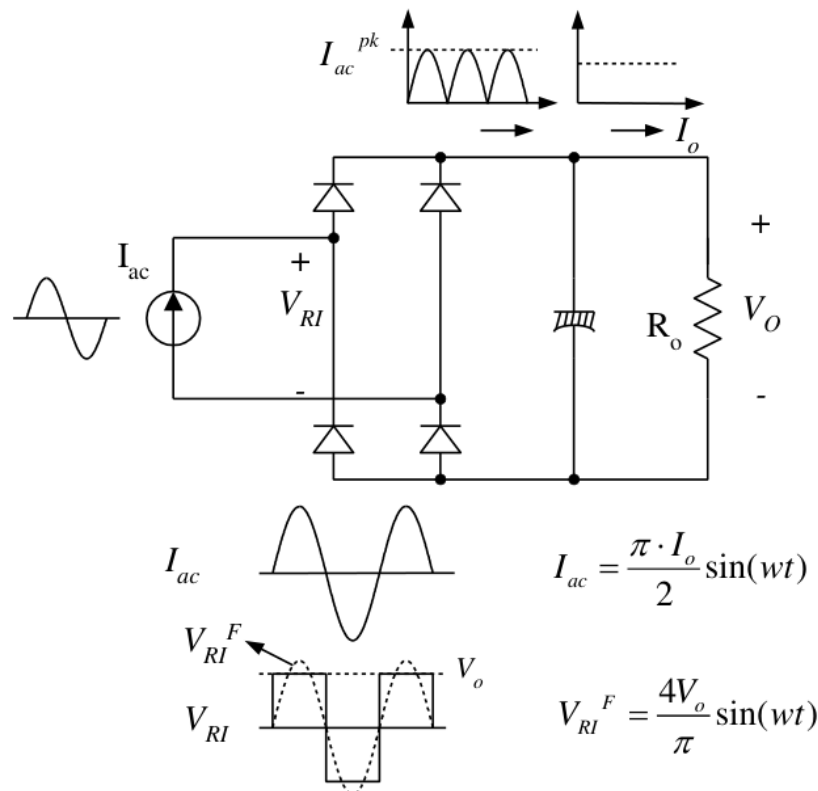


Figure 3: Why the rectifier and the load collapse into one resistance. The rectifier draws a square current wave and the output is stiff, so on a fundamental basis the whole of it looks resistive. (ON Semiconductor AN-4151.)

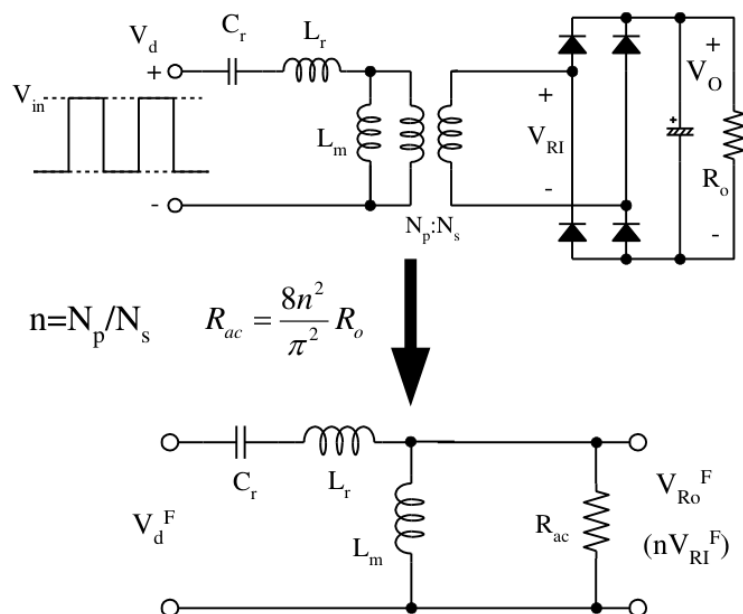


Figure 4: The result: one ac network. Its voltage transfer is the gain M and its damping is Q . The R_{ac} written here uses the average output power, which is the only power a two-stage converter has. Put the line-peak power into the same expression instead and the result halves; Section 6.4 says why. (ON Semiconductor AN-4151.)

- **Refer the secondary to the primary.** The ideal transformer disappears and the load resistance is multiplied by n^2 .

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- **Replace the rectifier and the output filter by one resistor.** The rectifier draws a square current wave and the output is stiff, so on a fundamental basis the whole of it looks resistive.
- **Keep only the fundamental of the drive.** The tank is a filter centred near the switching frequency, so the harmonics of the square wave contribute little. This is *first harmonic approximation*, and it is why every LLC design ends with a simulation or a measurement rather than with the equations.

What is left is one ac network, and it is a voltage divider whose ratio changes with frequency. Two numbers describe it and both are used on every page from here on:

- **the gain M** — the voltage across R_{ac} divided by the fundamental driving it. Because C_r and L_r cancel at the series resonance f_r , **$M = 1$ there whatever the load.** Below f_r the network can step up ($M > 1$) and above it steps down.
- **the quality factor $Q = Z_0/R_{ac}$** , with $Z_0 = \sqrt{L_r/C_r}$ the characteristic impedance of the tank. Q says how heavily the tank is loaded: **Q rises with output power.** At $Q = 0$ (no load) the gain curve is tall and peaked; loading it pulls the peak down and flattens the curve.

Frequency is always written normalised, $f_n = f_{sw}/f_r$, so that one family of curves serves every tank. $M(f_n, Q)$ always means the gain of the right-hand circuit above.

2.5 The two resonances

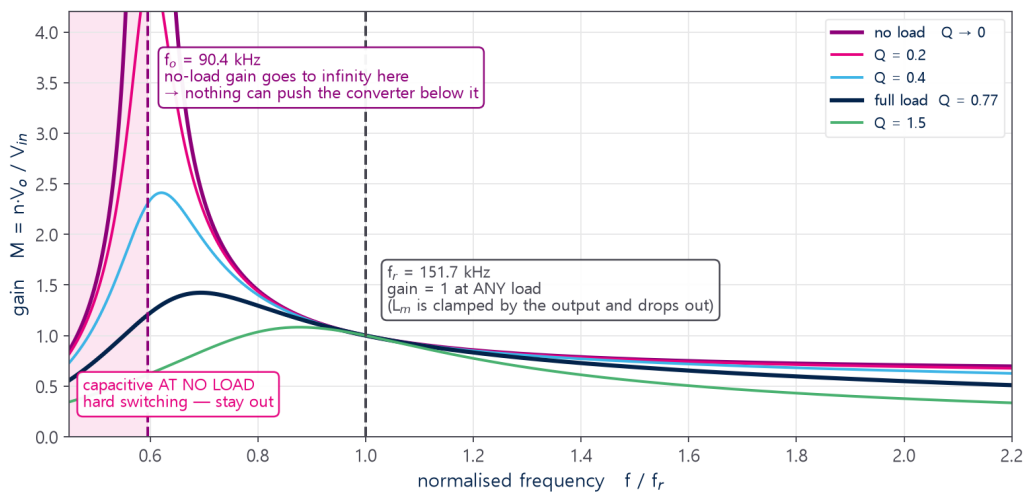
Every LLC has two. When the secondary conducts, the reflected output voltage clamps L_m and it drops out of the circuit, leaving the series resonance

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad (1)$$

At no load nothing clamps L_m , and the tank resonates with both inductances:

$$f_o = \frac{1}{2\pi\sqrt{(L_r + L_m) C_r}} \quad (2)$$

At f_r the gain is exactly 1 for any load, because the load term vanishes. At f_o the no-load gain is without limit. A single-stage converter spends most of the line cycle in the band between them — here 90.4 to 151.7 kHz — while a two-stage LLC sits close to f_r nearly all the time.



Between f_o and f_r the tank can boost, and that band is where an LLC lives. Follow one loaded curve down, though, and the gain stops rising at its own peak and falls again - the shaded edge is the no-load position of that peak, not the loaded one.

Figure 5: The two resonances, and the band a single-stage converter lives in.

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Two expressions are not much use without knowing what each one does to the current, and the current is where the difference is visible.

Before the waveforms, the two things the circuit can be doing. Every operating mode is one of these or both of them in sequence, and the currents follow from which one is running.

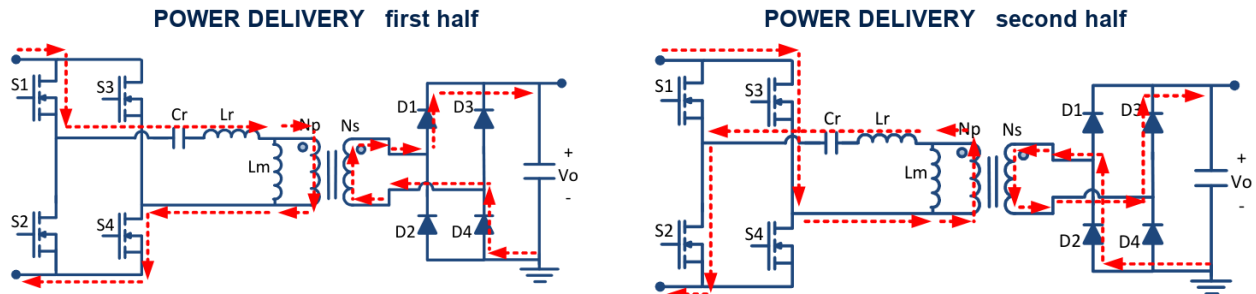


Figure 6: Power delivery. The tank is excited by the bridge and the resonant current exceeds the magnetising current, so the difference passes through the transformer to the rectifier and the load. The voltage across L_m is the reflected output, so L_m is clamped and takes no part in the resonance — which is why this interval resonates at f_r . (Infineon AN 2012-09, headings added.)

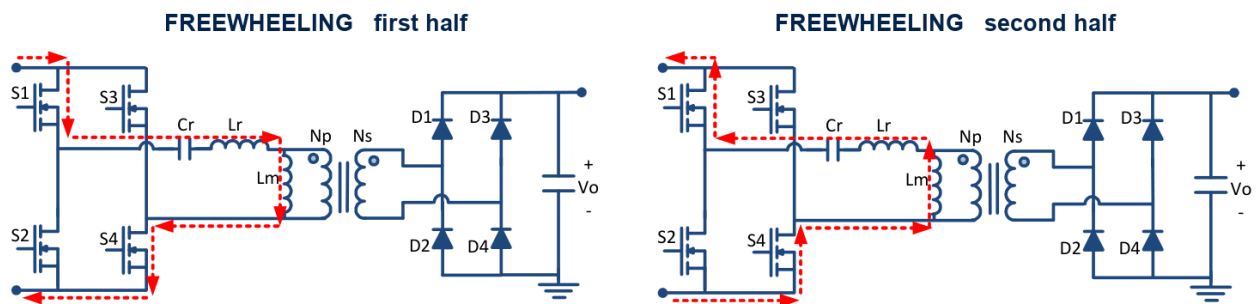


Figure 7: Freewheeling. The resonant current has fallen to the magnetising current, so nothing is left to pass to the secondary and the rectifiers are off. With the secondary disconnected L_m is no longer clamped and joins the resonance — this interval rings at f_o . (Infineon AN 2012-09, headings added.)

Those two circuits are the two resonances. The expressions at the top of this section are not abstractions: each one is the tank of one of these two pictures. Which one is running, and for how long, is the whole of what follows.

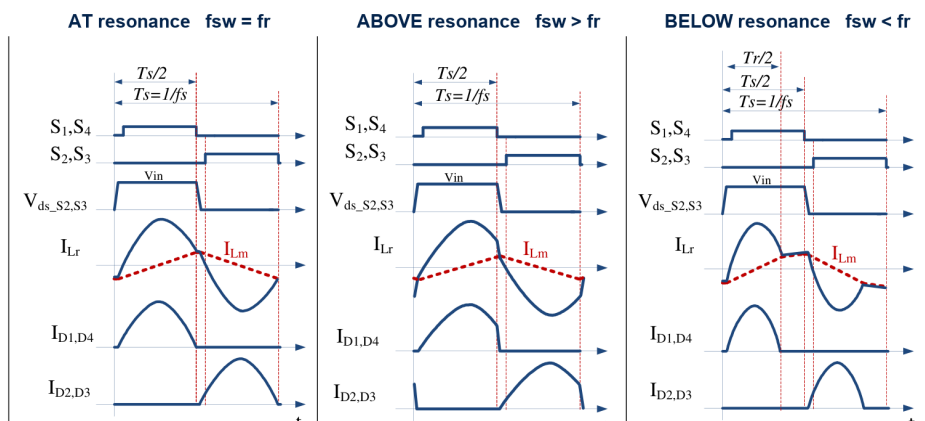


Figure 8: The three cases side by side. Gate signals, the drain voltage, the resonant current I_{Lr} with the magnetising current I_{Lm} dashed over it, and the current in each pair of rectifiers. Note $T_r/2$ and $T_s/2$ marked on the third panel: that gap is the whole story. (Infineon AN 2012-09, column headings added.)

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- **At f_r (left).** The resonant half period and the switching half period are the same length. I_{Lr} meets I_{Lm} exactly as the bridge changes and the rectifier current reaches zero exactly then too. Nothing is wasted at either end — this is the most efficient point an LLC has, and it is where a conventional design puts its nominal input.
- **Above f_r (centre).** The switching half period ends first, so the resonant half sine is **cut off part way**: I_{Lr} is still well above I_{Lm} when the bridge changes. The primary switches turn off more current, and the rectifier is interrupted while still conducting — hard commutation on the secondary.
- **Below f_r (right).** The resonant half period is the shorter one, marked $T_r/2$ against $T_s/2$. I_{Lr} finishes its half sine early and **lands on I_{Lm}** ; from that instant the two are one current, the rectifier current has already fallen to zero, and no power crosses to the secondary. That interval is called freewheeling.

That freewheeling interval is what f_o is. While it lasts the secondary is not conducting, so nothing clamps L_m and the tank is $L_r + L_m$ in series with C_r — it is ringing, at f_o . It looks flat only because f_o is so much lower than f_r that a fraction of its period is very nearly a straight line. Lower the switching frequency towards f_o and that interval grows until it is the whole half period, at which point the flat part is visibly a piece of a slow sine and the converter has stopped delivering power at all.

The two frequencies are not two operating points; they are the two ends of one. f_r is where the flat interval has just vanished. f_o is where it has taken over completely. Everything useful happens between them, and the fraction of the half period that is flat is, in practice, how much the tank is boosting.

Symbols are not standard across the literature. The borrowed figures have been relabelled to match, and each caption says so. Going the other way, to a manufacturer's application note, check first which frequency that document calls f_o — at least one widely used guide gives that name to the series resonance, the opposite of the meaning here.

A conventional LLC picks one side and stays there — usually just above f_r at nominal input, where the gain is close to 1 and the circulating current is smallest. Which side a *single-stage* converter is on is a different question, and it is answered in Section 4.5 once the rest of the machinery is in place.

2.6 The gain function and the three regions

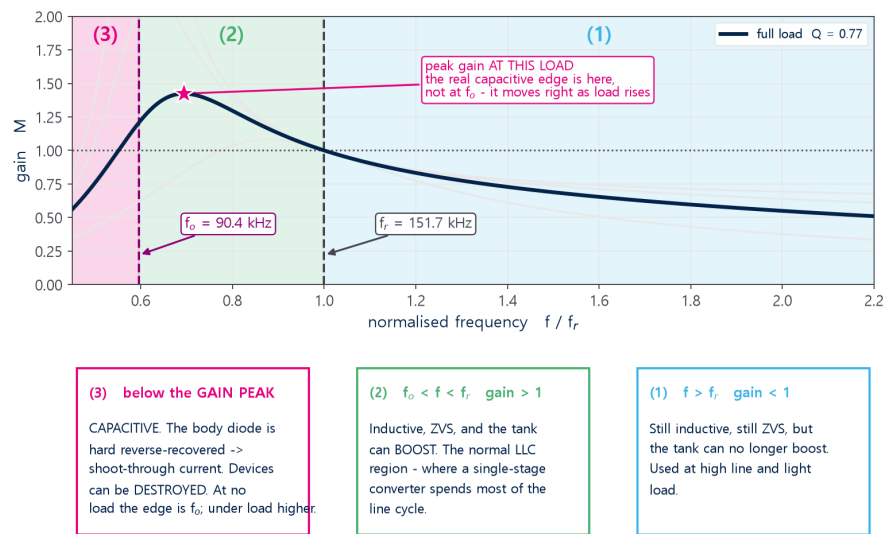
With the usual first-harmonic approximation the tank gain is

$$M(f_n, Q, \lambda) = \frac{1}{\sqrt{\left(1 + \lambda - \frac{\lambda}{f_n^2}\right)^2 + Q^2 \left(f_n - \frac{1}{f_n}\right)^2}} \quad (3)$$

with f_n , Q and λ as defined in the conventions. Two checks are worth making on it by eye before trusting any curve drawn from it. At $f_n = 1$ the second term under the root vanishes and the first becomes 1, so **$M = 1$ whatever Q** — that is the series resonance, and it is why every gain curve passes through the same point. **At no load** — $Q = 0$, so the second term is gone — raising f_n without limit walks M down to $1/(1+\lambda)$ and no further, which is the floor that Section 4.8 is about. Under load the Q term grows with f_n and the gain keeps falling towards zero instead.

The curve has three regions. Below the gain peak the tank is capacitive and the bridge hard switches, which must never be allowed; between the peak and f_r it is inductive and boosts; above f_r it is inductive and bucks. The two boundaries are drawn here at f_o and f_r , which is exact only at no load — the capacitive edge moves up with load, and the next section is about that.

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Frequency is the only control knob: raise it to cut the gain, lower it to raise the gain - but never below the gain peak. f_o is that limit only at no load. (region framing after ROHM TechWeb)

Figure 9: The three operating regions. Only the inductive ones are usable, and the boundary moves with load.

2.7 Capacitive and inductive, and why those words

The names have nothing to do with which component dominates. They describe **what the bridge sees**: the phase of the tank input current relative to the square wave driving it. The tank is a series network, so at low frequency C_r dominates and the current *leads* — the bridge is driving something that behaves like a capacitor. At high frequency the inductances dominate and the current *lags* — the bridge is driving something that behaves like an inductor. The word describes the load the bridge works into, not a part.

That phase is the whole of soft switching, which is why the distinction matters more than it sounds:

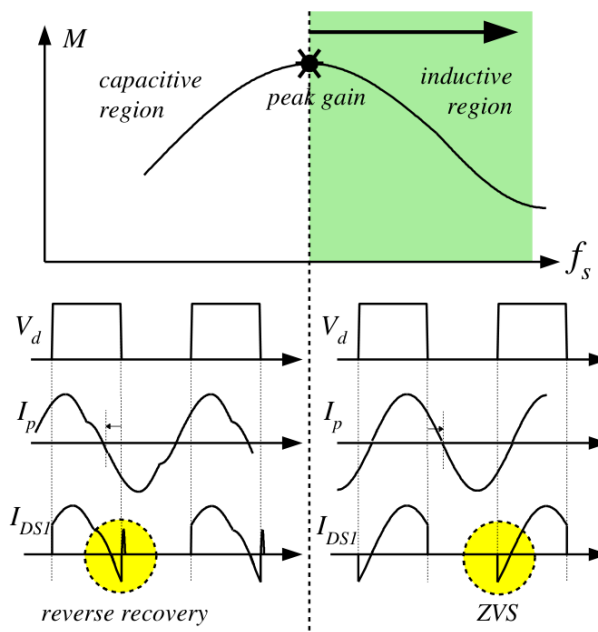


Figure 10: The same converter either side of the peak gain, and the waveforms that tell them apart. Above, where the boundary sits on the gain curve. Below, the bridge voltage V_d , the tank current I_p and the current in one switch I_{DS1} . On the left the tank is capacitive and the switch current shows a reverse-recovery spike; on the right it is inductive and the same instant is a clean ZVS transition. (ON Semiconductor AN-4151.)

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- **Inductive: current lags.** When a switch turns off, the current is still flowing in the direction that pushes the bridge node towards the other rail. The dead time lets it do exactly that, and the next device turns on at zero volts. **ZVS.**
- **Capacitive: current leads.** By the time the switch turns off the current has already reversed, so it pushes the node *back* where it came from. The next device then turns on into the full rail, and worse, the body diode of the device that was conducting is forced out with reverse recovery through a low impedance. **Hard switching, and the failure mode that destroys parts.**

This is why the capacitive region is not a performance trade-off to be balanced against something else. It is a boundary the design stays on one side of, and the controller carries a dedicated protection for the case where it does not.

2.8 The two boundaries are not the same boundary

This is the point most easily confused, because both pairs of words describe positions on the same axis. They are different lines on it.

	capacitive / inductive	below / above resonance
the boundary	the peak-gain frequency	f_r , the series resonance
does it move?	yes — with load	no, it is fixed by L_r and C_r
what it decides	ZVS or hard switching, on the primary	boost or buck, and ZCS or not, on the secondary
which side is allowed	inductive only, always	both sides are legitimate

Table 1: Two classifications, two boundaries, two consequences.

The peak-gain frequency sits between f_o and f_r , and it **risks as the converter is loaded**. At no load it coincides with f_o ; at full load it has moved some way up towards f_r . So the capacitive region is not a fixed band below f_o — it grows as load is added, and a frequency that is safely inductive at light load can be capacitive at overload.

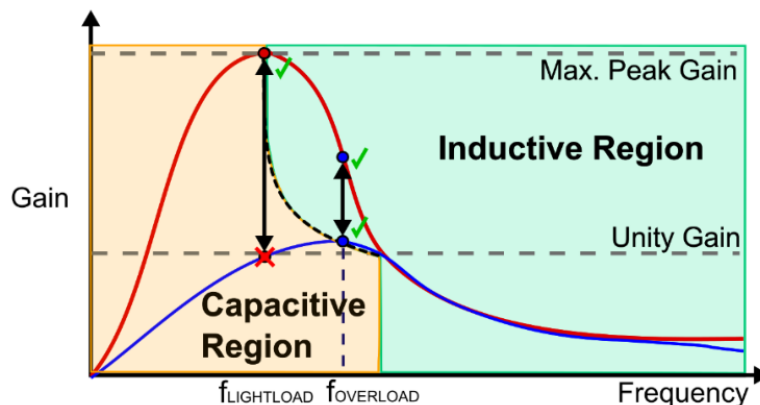


Figure 11: The capacitive boundary is not fixed. Loading the converter pushes the gain peak, and with it the edge of the capacitive region, to a higher frequency. (Monolithic Power Systems, Understanding LLC Operation part 2.)

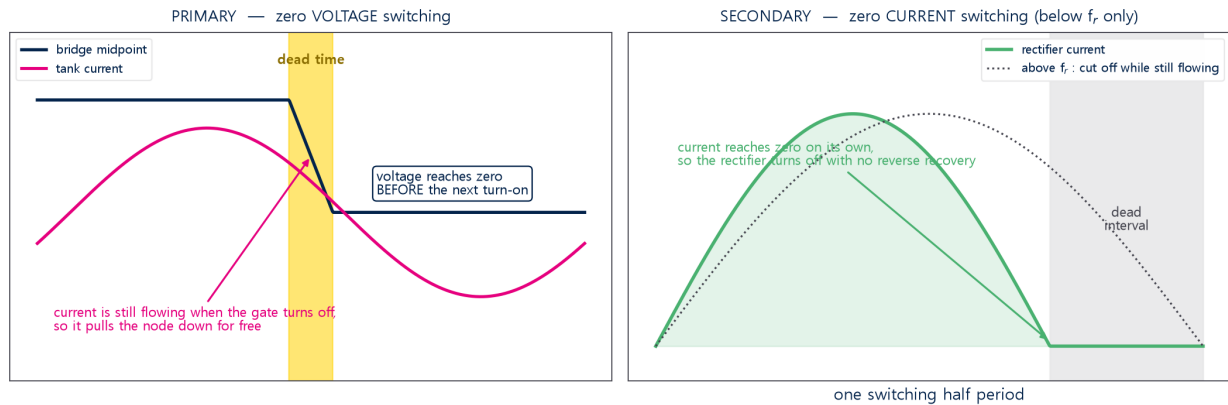
- A converter can be **below f_r and still inductive** — that is the ordinary boosting region, and it is where an LLC spends most of its life.
- It can be **below f_r and capacitive**, which is the fault. Going below f_r is not the danger; going below the peak is.
- It cannot be **above f_r and capacitive**, because the peak is always below f_r . Everything above f_r is inductive at every load.

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What this means for the design rule. Keep the oscillator floor f_{Min} above f_o . That is the *no-load* position of the boundary, so it is the weakest form of the requirement: necessary, but not enough on its own. The real check is the ZVS sweep in Section 6.5, which is run at load and at every phase of the line cycle, and the controller carries an anti-capacitive protection behind that.

2.9 ZVS and ZCS are not the same thing

Both are ways of switching a device while one of its two quantities is zero, so that their product — the loss — is zero. They apply to different devices here, for different reasons, and only one of them is guaranteed.



The two are different mechanisms on different devices. ZVS on the primary needs inductive operation and is required everywhere, and is lost the moment the converter crosses above it. ZCS on the secondary happens only below resonance, and is lost the moment the converter crosses above it.

Figure 12: Zero-voltage switching on the primary and zero-current switching on the secondary. Different devices, different mechanisms, different conditions.

- **ZVS — zero-voltage switching, on the primary switches.** The device turns on with no voltage across it. It works because the tank current is still flowing when the previous device turns off, and that current discharges the bridge node during the dead time. It needs **inductive operation**, which is to say anywhere above the gain peak — so it is available on both sides of f_r , and f_o is that limit only at no load. It is required everywhere, and Section 6.5 says how much margin this design has.
- **ZCS — zero-current switching, on the secondary rectifiers.** The device turns off with no current through it, so a diode is not forced out of conduction and a synchronous rectifier has no reverse recovery. It happens because the resonant current reaches zero on its own — which only occurs **below f_r** . Above f_r the rectifier is cut off while still conducting and reverse recovery comes back.

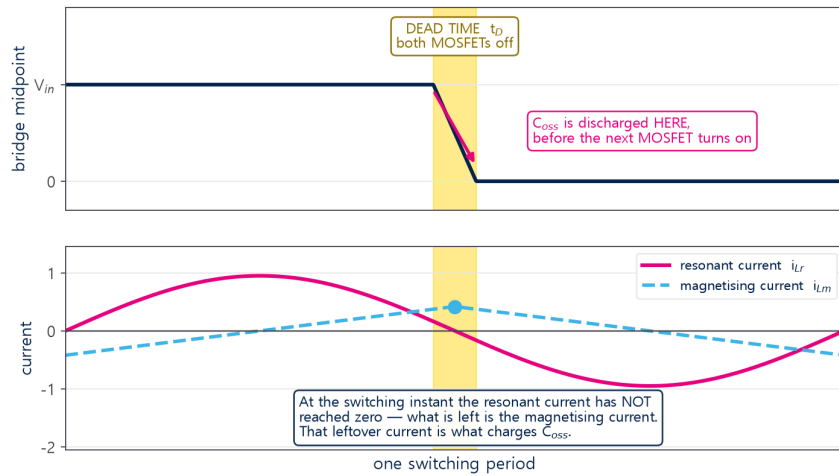
Which explains the previous figure. At low line the converter is below f_r for the whole cycle and gets ZCS on the secondary for free. At high line it is above f_r for much of the cycle and does not. That is a reason to care about the secondary body diode and the synchronous-rectifier dead time in a single-stage design, where a conventional LLC could ignore it.

2.10 Sizing the dead time so that ZVS actually happens

ZVS is what makes an LLC efficient, and it is a charge problem rather than a voltage problem. During the dead time the magnetising current must move enough charge to swing the bridge node across the rail before the opposite device turns on. The design quantity is the time the tank current takes to reach zero after the bridge transition, T_{ZC} , and the requirement is simply that it outlast the dead time:

$$T_{\text{ZC}} > t_D \quad (4)$$

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Two conditions, both needed: current must still be flowing at turn-off ($T_{ZC} > t_D$), AND the transition must finish inside the dead time ($t_D > T_T$).

Figure 13: How ZVS happens: the magnetising current, not the load current, discharges the bridge node during the dead time.

Use the output charge, not the flat-band capacitance. A MOSFET datasheet quotes C_{oss} , $C_{o(er)}$ and $C_{o(tr)}$, and they differ by three to five times. ZVS is a charge question, so $C_{o(tr)}$ (equivalently Q_{oss}) is the one to use. Sizing the dead time from the headline C_{oss} is optimistic by a large factor.

2.11 How an LLC is normally designed

The sequence below is the standard one, and every step exists because of something already established. It is worth having in mind before Section 6, which follows the same order and diverges from it in exactly two places.

- **1 Turns ratio from the nominal point.** Choose n so that the converter runs at or near $M = 1$ at nominal input, that is at f_r . That is where the circulating current is lowest and where the converter should spend most of its life.
- **2 The gain range the tank must cover.** From the input range and the output tolerance, M_{min} and M_{max} . M_{max} is asked at minimum input and full load, M_{min} at maximum input and minimum load.
- **3 Load resistance referred to the primary, R_{ac} ,** and with it the meaning of Q at full load.
- **4 m (equivalently λ) and Q , together, from the peak-gain chart.** These are not independent: for each m there is a curve of attainable peak gain against Q , and the design has to sit under the curve with margin. A larger m gives less circulating current and less peak gain; a smaller m the reverse.
- **5 The component values.** Q and R_{ac} give Z_0 , and Z_0 with the chosen f_r gives C_r and L_r ; m then gives L_m .
- **6 Verify.** That the gain at the worst corner is still under the peak with margin, that ZVS holds at the worst point rather than the nominal one, and that the currents and the flux are ones real parts can carry.

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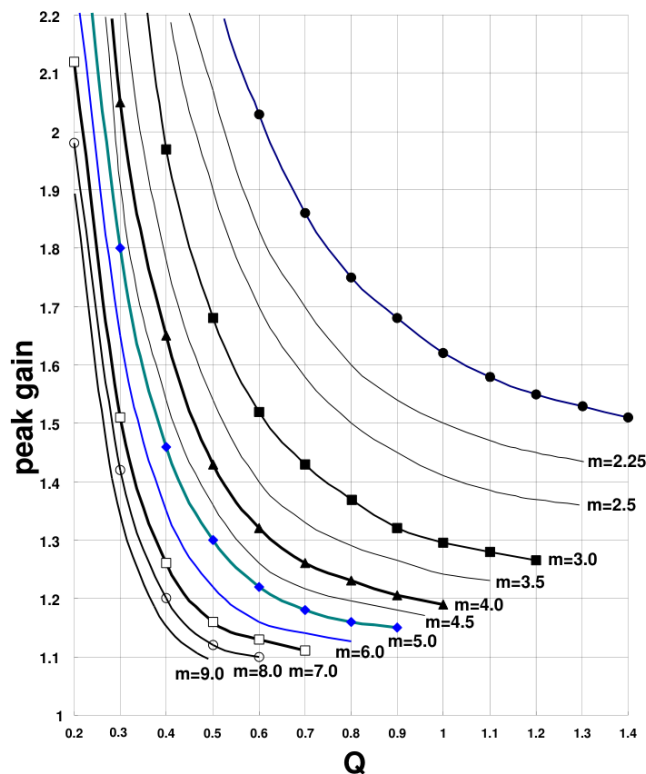


Figure 14: Step 4, and the reason m and Q cannot be chosen separately: the peak gain a tank can reach depends on both. Picking a required gain and a Q leaves only a band of usable m . (ON Semiconductor AN-4151.)

Two of those steps are the ones a single-stage converter breaks. Step 1 has no nominal point to sit at, because the input sweeps a half sine; and step 2 asks for a gain range that no single tank can cover, which is why the bridge itself has to change. Sections 4.1 and 5 are those two departures.

That is the first half. Everything above is the LLC on its own: a tank, a gain curve, and a frequency that moves the operating point along it. Nothing in it has anything to do with the mains. The rest of this section is the other half — what a power factor corrector is for and how the ordinary one works — and it can be read without reference to any of the above. Section 3 is where the two are put together.

2.12 What power factor correction is

Start with what happens if nothing is done. A bridge rectifier feeding a capacitor holds the bus near the peak of the mains, so the diodes can only conduct during the short window in which the mains is above that. All of the charge the load will use over the whole cycle has to arrive inside that window.

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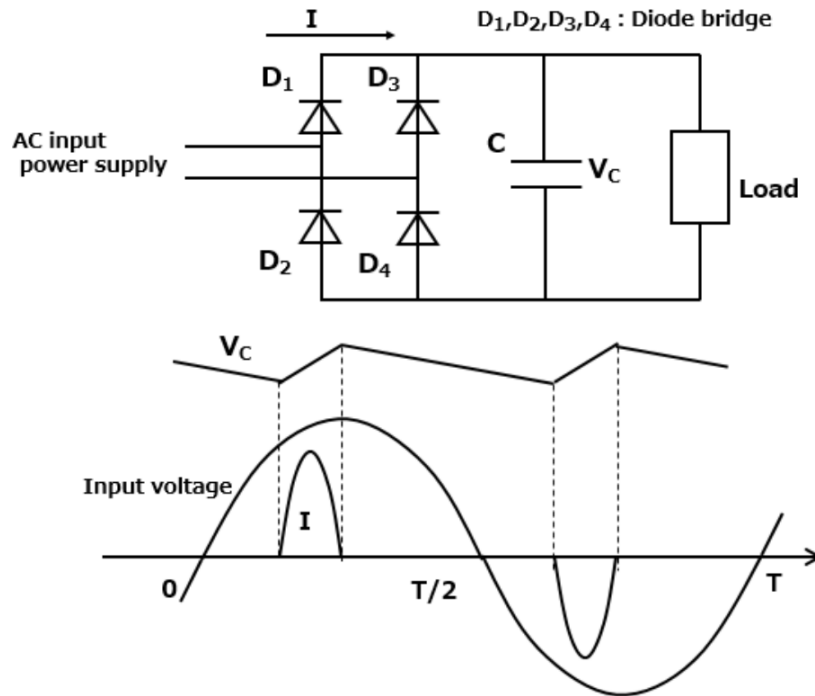


Figure 15: A capacitor-input rectifier and what it draws. V_C stays near the peak, so the diodes conduct only in the two narrow windows where the mains is above it, and the current I inside them is correspondingly tall. (Toshiba, Power Factor Correction Circuits.)

The consequences are all consequences of that one fact. The peak current is several times what a resistor of the same average power would draw, so the wiring, the fuse, the bridge and the source all have to be sized for it. And a pulse train that narrow is mostly harmonics.

2.13 Power factor, and why it is not the same as distortion

Power factor is the ratio of real power to apparent power:

$$PF = \frac{P}{V_{rms} I_{rms}} = \cos \varphi_1 \times \frac{I_{1,rms}}{I_{rms}} \quad (5)$$

It has two independent factors and they fail for different reasons. **Displacement** is the phase between the voltage and the fundamental of the current, and it is what a motor gets wrong. **Distortion** is how much of the rms current is at the fundamental at all, and it is what a rectifier gets wrong. The current in the figure above is roughly in phase, so its displacement factor is near 1 — and it still scores about 0.6, because the rest of the rms is harmonics.

The related figure of merit is **total harmonic distortion**, the harmonic content as a fraction of the fundamental:

$$THD = \frac{\sqrt{I_{rms}^2 - I_{1,rms}^2}}{I_{1,rms}}, \quad \frac{I_{1,rms}}{I_{rms}} = \frac{1}{\sqrt{1 + THD^2}} \quad (6)$$

The two are not interchangeable. A converter can hold PF above 0.99 and still fail a harmonic limit, because the standard limits *individual* harmonics in amperes rather than a single ratio. Harmonics do not deliver power; they heat the neutral, saturate distribution transformers and disturb other equipment on the same supply, which is why they are regulated. IEC 61000-3-2 sets the limits, and which limit applies depends on the class: a mains rectifier of this shape is Class D, exempt below 75 W and held to a milliamp-per-watt line above it. **But Class D only reaches 600 W** — this design draws 685 W at the input, above that, so the absolute Class A limits apply instead. Settle the class first.

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2.14 How a corrector fixes it

Force the input current to follow the input voltage and the converter looks like a resistor to the mains: both factors go to one at once. The standard implementation is a **boost converter placed straight after the bridge**, switching fast enough that the mains looks frozen within one switching cycle.

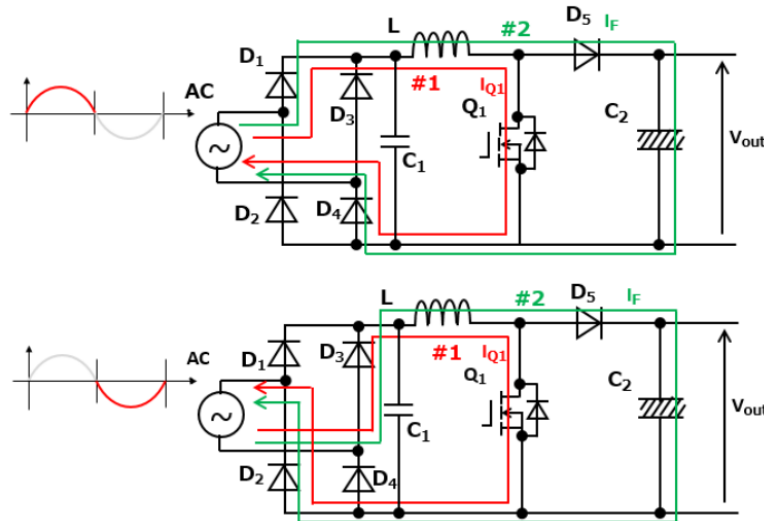


Figure 16: A boost corrector, with the current path traced for each half of the line cycle: red while the switch is on and the reactor charges, green while it is off and the reactor delivers to the bus. (Toshiba, Power Factor Correction Circuits.)

- **Why boost.** The input is a rectified sine that passes through zero, so the stage has to be able to step up by an unlimited ratio near the crossing — only boost can. Its input current is also continuous, which is what makes shaping it possible at all.
- **The inner loop** regulates inductor current to a reference. It has to be fast compared with the switching frequency.
- **The outer loop** regulates the bus voltage, and is deliberately made **slower than $2f_i$** . If it were fast it would fight the ripple on the bus, modulate the level within the line cycle and flatten the very shape the corrector exists to make. A slow voltage loop is not a compromise here; it is a requirement.
- **The multiplier** joins them: shape from the mains, level from the output. That is the whole of the control law.

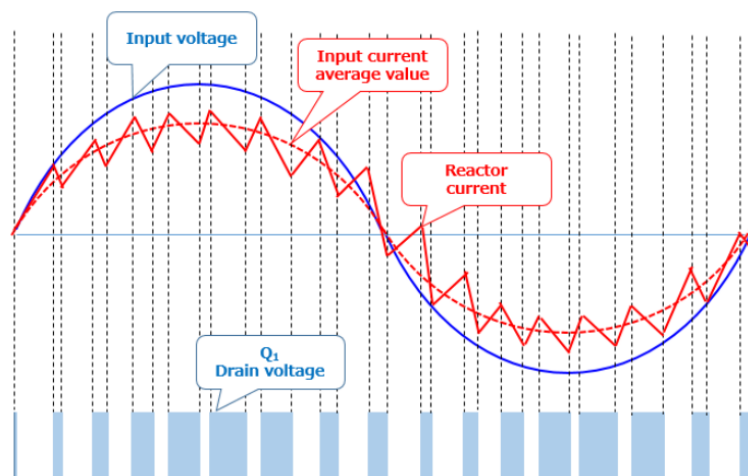


Figure 17: The result, in continuous conduction mode. Switching ripple rides on the reactor current; its average follows the input voltage, which is what the two loops were arranged to produce. (Toshiba, Power Factor Correction Circuits.)

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This comes back later. The same conflict reappears in the single-stage converter, in a sharper form: there the outer loop is the *only* loop, its output is a power command, and any $2f_l$ ripple that survives it becomes input current distortion directly. That is Section 6.19.

The corrector delivers what was asked for, and hands over a new problem in doing so. A current that follows the voltage means an input power that follows $\sin^2\theta$: zero twice per cycle, twice the average at the crests. The load still wants constant power. **Something has to store the difference**, and where that something sits is the whole of the next section.

2.15 Why they are normally two stages

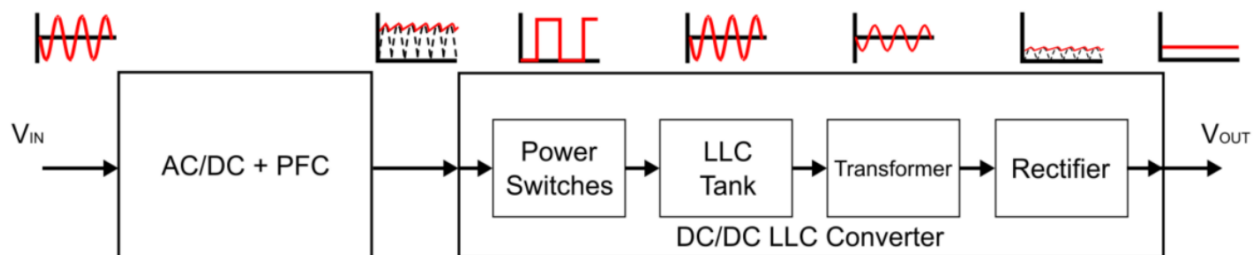


Figure 18: The usual two-stage arrangement, with the waveform at every node. The correction happens in the first block and the LLC works from the dc bus it produces. (Monolithic Power Systems, Understanding LLC Operation part 2.)

The bus capacitor between a boost PFC and an LLC does two jobs at once, and they are usually confused with each other:

- it **buffers** the difference between the pulsating input power and the constant output power;
- it **isolates** the LLC from the mains waveform, so the tank sees a nearly constant input and has to cover only a narrow gain range.

Removing the boost stage removes both services at once. The buffering has to be done somewhere else, and the tank now has to work directly from a rectified sine that goes to zero a hundred times a second. What that costs, and why it works anyway, is the rest of the story.

3 Why single stage, and what it costs

3.1 The energy a unity power factor cannot deliver

Drawing a sinusoidal current in phase with a sinusoidal voltage makes the instantaneous input power a raised sine squared. It swings between zero and twice the average, at twice the line frequency:

$$p_{in}(t) = V_{ac}I_{ac} [1 - \cos(2\omega_l t)] \quad (7)$$

The load wants constant power. The difference has to be stored and returned a few milliseconds later. This requirement does not depend on topology; it follows from asking for unity power factor at all. Figure 19 shows the imbalance that must be buffered.

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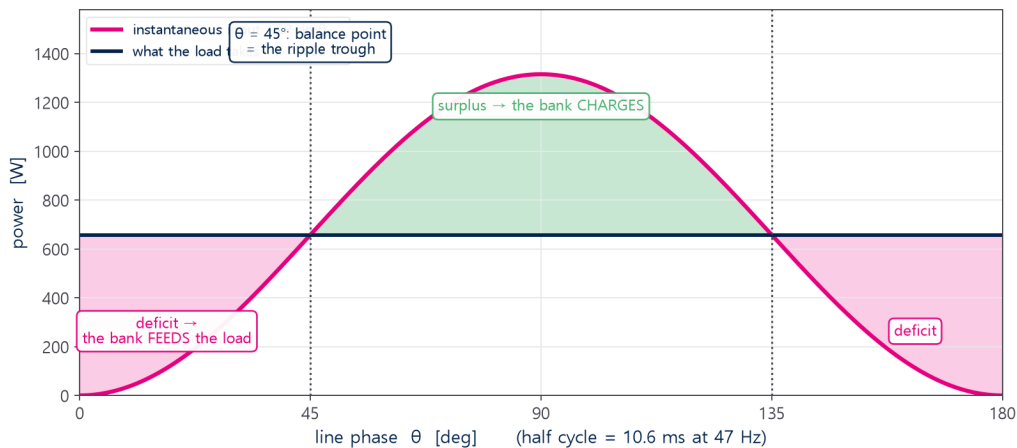


Figure 19: Unity power factor forces a $2f_l$ energy imbalance. The shaded areas are what a capacitor somewhere must absorb and give back.

3.2 Moving the buffer from 400 V to the output

In the two-stage converter that buffer is the bus capacitor, at 400 V. In the single-stage converter there is no bus, so the buffer moves to the output. **The joules do not change — the voltage they sit at does**, and a capacitor only ever returns the energy between its starting voltage and the lowest the load will accept:

$$E = \frac{1}{2}C(V^2 - V_{min}^2) \quad (8)$$

Hold-up is the clearest way to see the price, because both architectures must ride out the same 12 ms at full power — 7.9 J either way. On a 400 V bus falling to 320 V that is a 274 μ F part. On a 25 V output falling to 19 V it is **67.2 mF**: the same energy, 245 times the capacitance, because the usable voltage window shrank from $400^2 - 320^2$ to $25^2 - 19^2$. That single ratio is the price of the architecture. It should be settled before anything else is designed: if the enclosure cannot house the bank, the output voltage is what has to change.

The bank finally built is larger still. 67.2 mF only satisfies hold-up. Ripple asks for 71.2 mF on the same output and wins, so the design carries **75.2 mF**. Which of the two decides is a question about the specification, and Section 6.16 settles it.

The trade in one line. Removing the boost stage removes a switch, an inductor, a diode and a 400 V electrolytic, and their losses with them. It adds a very large low-voltage capacitor bank, a $2f_l$ ripple on the output that the load must tolerate, and a resonant tank that has to work over a much wider range. Whether that is a good trade depends almost entirely on how much output ripple the load will accept.

3.3 What the load must tolerate

Because the buffer is on the output, the output carries the $2f_l$ ripple directly; there is no second stage to reject it. The ripple is set by the bank and the load, not by the control loop, and it cannot be regulated away — attempting that puts distortion on the input current instead (Section 6.19). Here the specification is **5 % peak-to-peak**, which is 1.18 V on a 25 V rail. **How tight that figure has to be is a property of the load, not of the converter**, and it has to come from the specification rather than from a design rule.

4 Operating principle

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4.1 Gain is a boundary condition, not a control variable

In a two-stage LLC the controller commands gain: it moves the switching frequency until the output is right. In a single-stage PFC LLC it cannot, because **both ports are voltage sources**. The output is a millifarad-class capacitor whose voltage cannot move within a switching cycle, and the input is the rectified mains, which the converter does not control either. The instantaneous gain is therefore forced:

$$M(\theta) = \frac{n V_{o, eff}}{V_{drive}(\theta)} = \frac{n V_{o, eff}}{\sqrt{2} V_{ac, eq} \sin \theta} \quad (9)$$

with θ the line phase angle. The reasoning is worth spelling out, because it is the hinge of the whole topology. A gain is a ratio of two voltages. In a two-stage converter the output voltage is free to move — the loop moves it — so commanding a gain is the same as commanding an output. Here **both of those voltages are already pinned by something outside the converter**: the input by the mains, the output by a bank so large that it cannot change within a switching cycle. Their ratio is therefore fixed too, moment by moment, and the controller has no say in it.

So what does moving the switching frequency do? It moves **Q** — the loading — and **Q** is power. The tank is told what gain to produce and is left to choose how much current it draws to produce it. **In this converter frequency is a power command, not a voltage command.**

4.2 Two divergences that cancel

Read that equation near the mains zero crossing and it looks impossible: as $\theta \rightarrow 0$ the required gain rises as $1/\sin\theta$ without limit. The converter survives because the load vanishes at the same time. A unity-power-factor input draws power proportional to $\sin^2\theta$, so

$$Q(\theta) = Q_{pk} \sin^2 \theta \quad (10)$$

and an unloaded LLC has without limit gain at its lower resonance f_o . Two divergences meet, and the operating point walks down towards f_o as the mains approaches zero (Figure 20). **f_o is therefore the frequency floor of the entire design**, and putting the oscillator clamp below it is one of the few mistakes that destroys hardware.

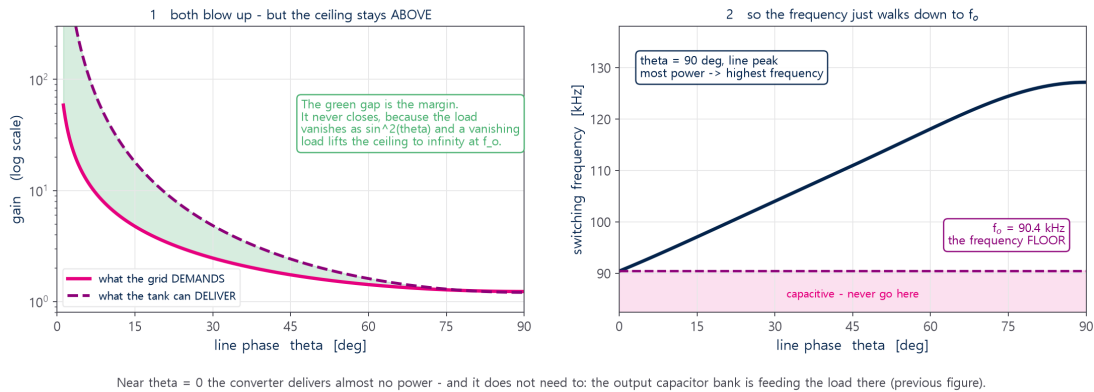
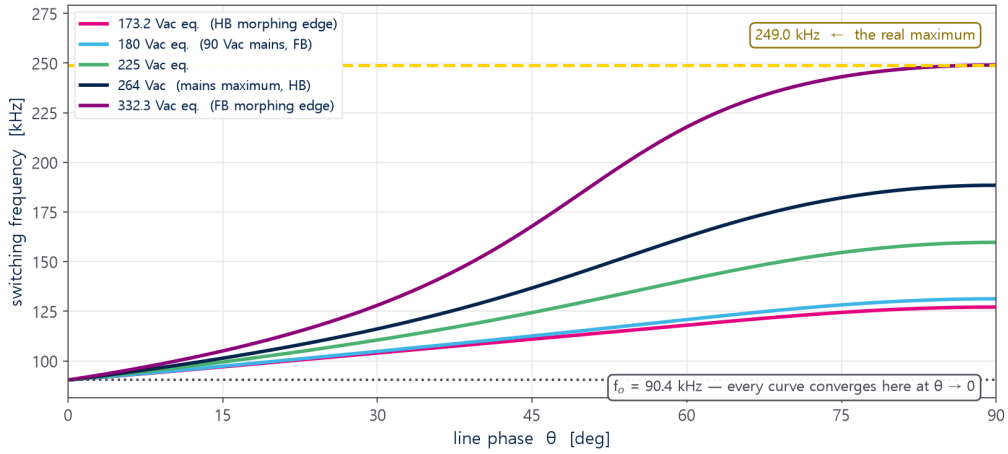


Figure 20: Near the zero crossing the required gain diverges and the load vanishes together, so the operating point converges on f_o .

4.3 Frequency modulation is the power factor correction

Put those together and the control law falls out. Over a line half cycle the controller sweeps the switching frequency so the power drawn follows $\sin^2\theta$. There is no current loop, no multiplier and no separate PFC stage: **the frequency profile $f_{sw}(\theta)$ is the power factor correction**. Figure 21 shows it for this design at both ends of the equivalent input range.

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The highest curve is NOT the maximum mains (264 V) — it is the FB morphing edge at 332 V equivalent. Miss morphing and you never plot that curve.

Figure 21: $f_{sw}(\theta)$ over a line half cycle. The excursion is the power factor correction; the floor is f_o .

Working that profile out looks like it needs a numerical root search. At each θ the gain equation has to be solved for f_n , and it has two roots: a capacitive one and an inductive one. **It does not.** Substituting $M_{req} = M_{pk}/\sin\theta$ and $Q = Q_{pk}\sin^2\theta$ and writing $x = 1/f_n^2$ turns the gain equation into a cubic in x :

$$\lambda^2 x^3 + (q - 2\lambda(1 + \lambda))x^2 + ((1 + \lambda)^2 - 2q - \frac{u}{M_{pk}^2})x + q = 0, \quad u = \sin^2 \theta, \quad q = Q_{pk}^2 u^2 \quad (11)$$

One root is always negative: the cubic is $+q$ at $x = 0$ and runs to $-\infty$ as x grows. The other two are the two crossings of one curve. On $x > 0$ the right-hand side of the gain equation starts at infinity, falls to a single minimum and rises again. So a required gain the tank can actually produce is met at **exactly two** positive values of x : the capacitive crossing and the inductive one. All three roots are then real. The physical, inductive root is the **middle** one, because $x = 1/f_n^2$ runs backwards against frequency. There is no branch to choose: in the trigonometric form of Cardano it is always the $k = 1$ branch. If the required gain is below that minimum there is no positive root at all, which is the no-solution case of Section 4.8 rather than a numerical failure.

Why this matters beyond elegance. A grid search needs an upper bound on f_n , and a bound chosen from the ac maximum rather than the morphing corner with no warning cannot represent the real operating point. The closed form has no grid, no bracket and no starting guess, so that failure mode does not exist. The inverse problem — the $\sin^2\theta$ that gives a required f_n — is a quadratic and is likewise exact.

The zero-crossing dead zone. Very near $\theta = 0$ the required gain exceeds anything the tank can produce and the converter simply stops drawing current. This is inherent to the topology and shows up as third-harmonic distortion on the input current. It is one of two mechanisms that set the achievable THD; the other is the voltage loop (Section 6.19).

4.4 The feedback pin is a power command

Combining the burst-mode expression in the datasheet with the multiplier in the block diagram gives the physical meaning of the feedback voltage. It is not an error signal in the usual sense:

$$V_{FB} = V_{os} + \frac{2K_{HV}}{K_{MKFF}} R_{CS} P_{in} = 0.5 \text{ V} + 0.167 \frac{\text{V}}{\Omega \cdot \text{W}} R_{CS} P_{in} \quad (12)$$

Substituting the 2.8 V feedback span gives $2.8/0.167 = 16.8 \Omega \cdot \text{W}$, which is exactly the maximum-power rule in the datasheet. **Maximum power limiting, burst entry and overload detection therefore all sit on one scale**, and

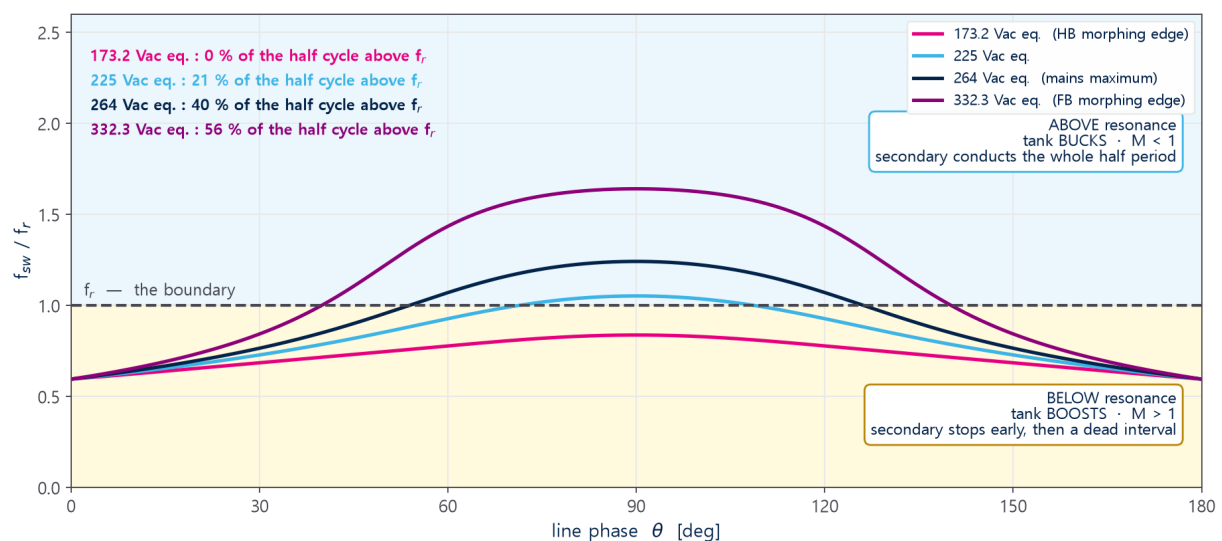
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that scale is R_{CS} . The outer optocoupler loop commands an *input power*; the internal loop distributes that power over the line cycle as $\sin^2\theta$.

This is why the current-sense pin tolerates no filter and no series resistor. The maximum-power law and the over-current threshold read the same pin, so anything that shifts it shifts both.

4.5 Which side of resonance this converter runs on

Section 2.5 left the question open. Now it can be answered, and the answer is that a single-stage converter does not pick a side. The equivalent input sweeps a half sine every 10 ms, the required gain sweeps with it, and the operating point walks across f_r and back within one line cycle. How much of the cycle it spends on each side depends on the mains voltage.



At low line the converter never leaves the boosting region. At the high morphing edge it crosses into the bucking region around the line peak and comes back. A single-stage converter therefore has to be designed for BOTH sides.

Figure 22: Which side of f_r the converter is on, over a line half cycle, at four equivalent inputs. At the low corner it never leaves the boosting region; at the high corner it spends most of the cycle bucking.

Both sides have to be designed for. The boosting side sets the gain requirement and the ZVS margin. The bucking side sets the top switching frequency, and it takes the secondary out of zero-current switching — so the rectifier body diode and the SR dead time matter here in a way they do not in a conventional LLC. A design checked only at the line peak, or only at low line, has looked at one of the two converters.

4.6 Why λ must be about 0.5

A classic LLC uses $m = L_p / L_r$ between 5 and 10, that is λ between 0.11 and 0.25. A single-stage converter cannot: it has to produce very high gain near the zero crossing, and peak gain falls as λ falls. This design runs $\lambda = 0.55$, that is $m = 2.82$. The evaluation board ST publishes for this part runs $\lambda = 0.500$, which is the strongest available confirmation that the value is structural and not a quirk of one design.

The price is circulating current. A small L_m means a large magnetising current that carries no power but does carry conduction loss, at every load. **A single-stage PFC LLC is less efficient than a two-stage LLC, and that is inherent rather than a design defect.**

4.7 $f_{sw,max}$ does not check the tank — it computes it

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Two of the four candidates for λ carry the specified maximum switching frequency in a denominator:

$$\lambda_2 = \frac{\lambda_1}{1 - \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad \lambda_{TD} = \frac{\lambda_1}{1 - \frac{\pi^2}{8} \left(\frac{f_r}{f_{sw,max}}\right)^2} \quad (13)$$

λ_1 is the plain minimum-gain condition. The other two are the same condition with the frequency ceiling folded in: the tank has to reach the lowest required gain *before* it runs out of frequency. Here $f/f_{sw,max} = 150/225$, so the squared ratio is 44.4 %, both denominators fall well below one, and the requirement rises from $\lambda_1 = 0.560$ to 1.009 and 1.241. **The number typed as $f_{sw,max}$ roughly doubles the λ the tank is asked for**, and as it approaches f_r the denominators go to zero and the requirement diverges.

So where does $f_{sw,max}$ come from? The common rule is $1.5 \times f_r$, which is what the 225 kHz here is. It is a **choice**, not a limit — and it sits upstream of the tank, so changing it changes L_m . That is the opposite of how a maximum usually behaves in a design sheet, and a round number can be typed into it without noticing what moves.

It should not be confused with f_{Max} , the oscillator ceiling. That one *is* a limit: R_T , C_T and the idle time fix it in silicon, the control loop cannot push past it however much gain it wants, and it is checked after the fact by k_{ceil} — 320 kHz against an operating maximum of 249 kHz, a factor of 1.28. The two sit at opposite ends of the design: one decides the tank, the other reports whether the oscillator can serve it.

4.8 The other bound on λ , and where it has no solution

The condition above sets a floor under λ . The ceiling comes from the opposite end: the tank must also be able to reach the *lowest* required gain, which happens at the highest equivalent input and no load. That is where the usual design rule without showing it breaks down, because the no-load gain of an LLC does not fall indefinitely. Raising the frequency only walks it down to an asymptote:

$$M_\infty = \lim_{f \rightarrow \infty} M_{no\ load} = \frac{1}{1 + \lambda} \quad (14)$$

If the required minimum gain lies below M_∞ , no frequency satisfies the condition at all. A spreadsheet solving for that frequency returns an error rather than a number, and the error is easy to mistake for a broken formula. It is not: the question simply has no answer.

This design is below the asymptote: $M_\infty = 0.6452$ against a requirement of 0.6409. There is no frequency at which the unloaded tank reaches the required gain, and the margin is thin enough — well under one per cent — that no full-load check would ever show it. Raising the turns ratio raises the requirement and brings the solution back.

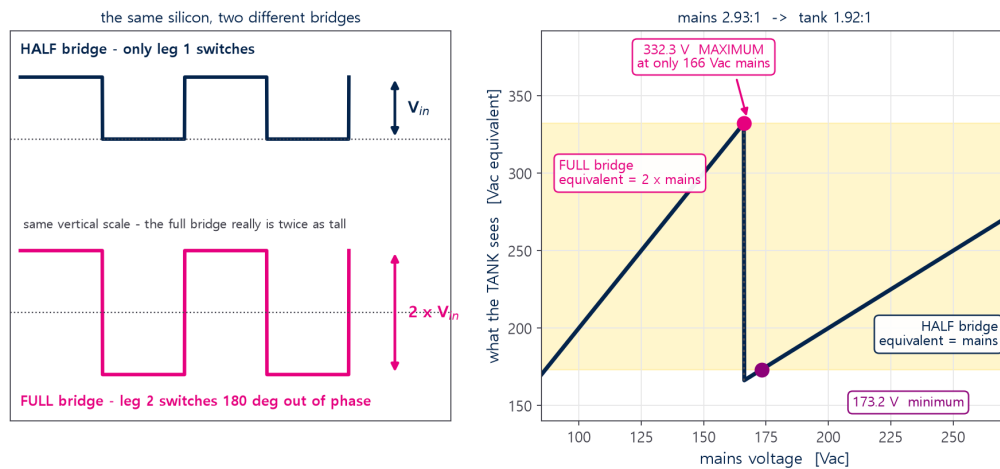
Losing the solution is not a failure. No load at high line is the region **burst mode** owns, not frequency control. Under load the gain curve falls further and the corner arrives sooner, so every full-load check is unaffected — which is exactly why this can go unnoticed. What is worth knowing is which of the candidate ratios still regulates on frequency alone, because no full-load number reveals it.

5 Topology morphing

A resonant tank has a usable gain range of roughly 1.5:1 at full load. A universal mains asks for 2.93:1. No single tank covers that and keeps ZVS, so the L6790A changes the bridge instead.

Below a mains peak of 235 V the part runs a **full bridge**, which drives the tank with twice the rail. Above 245 V_{pk} it runs a **half bridge**. The tank therefore sees 173.2 to 332.3 Vac equivalent, a range of 1.92:1, and one tank can cover it.

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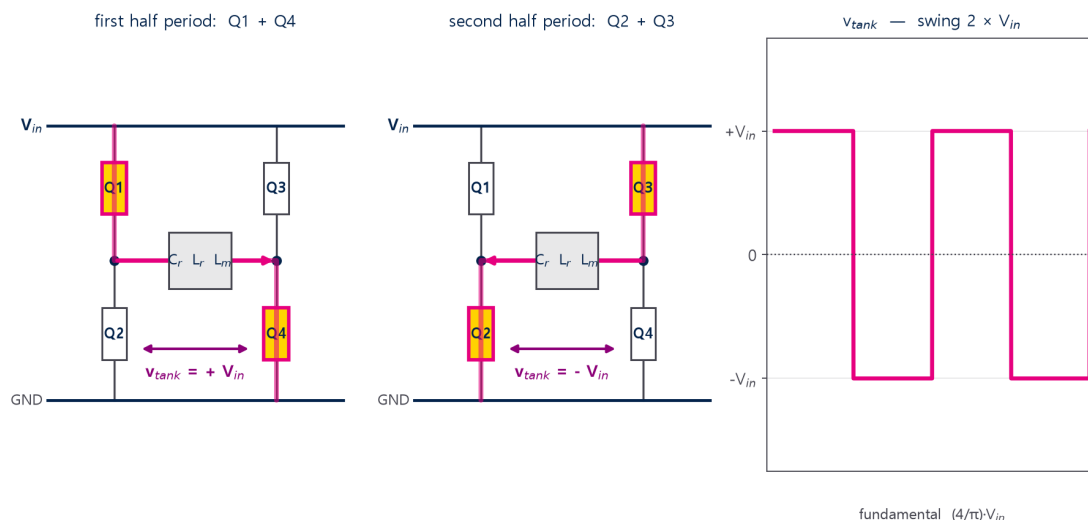


Half bridge is entered at 245 V peak going up and full bridge returns at 235 V peak coming down - 10 V of hysteresis - and the change is made by holding LOUT2 statically high, no extra hardware. The worst corners are these EDGES, not the ends of the mains range.

Figure 23: Morphing collapses a 2.93:1 mains range into a 1.92:1 range at the tank. The worst corners are the morphing edges, not the ends of the mains range.

5.1 Full bridge

Diagonal pairs conduct together. Q1 and Q4 put $+V_{in}$ across the tank, Q2 and Q3 put $-V_{in}$ across it, so the drive is a square wave of amplitude V_{in} and its fundamental is $(4/\pi)V_{in}$. All four devices switch, and each carries the tank current for half of every period.



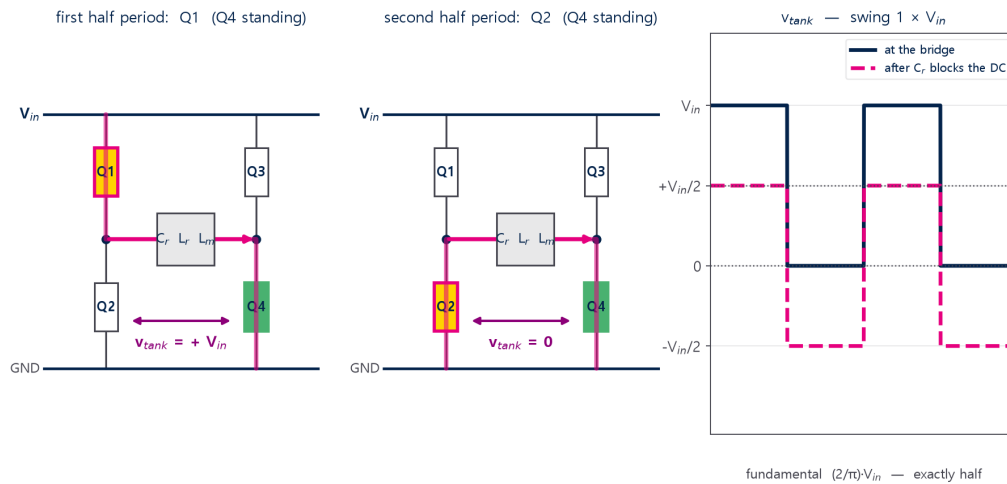
Diagonal pairs conduct together, so the tank is driven from $+V_{in}$ to $-V_{in}$ every cycle. Every one of the four devices switches, and each carries the tank current for half the period.

Figure 24: Full bridge: conduction path in each half period, and the resulting tank drive.

5.2 Half bridge

LOUT2 is held statically high. Q3 never turns on and Q4 never turns off, so the leg-2 midpoint becomes ground and the bridge output swings between 0 and V_{in} . C_r is already in series with the tank, so it blocks the $V_{in}/2$ of dc for free and the tank sees $\pm V_{in}/2$ — a fundamental of $(2/\pi)V_{in}$, exactly half. No extra hardware is involved.

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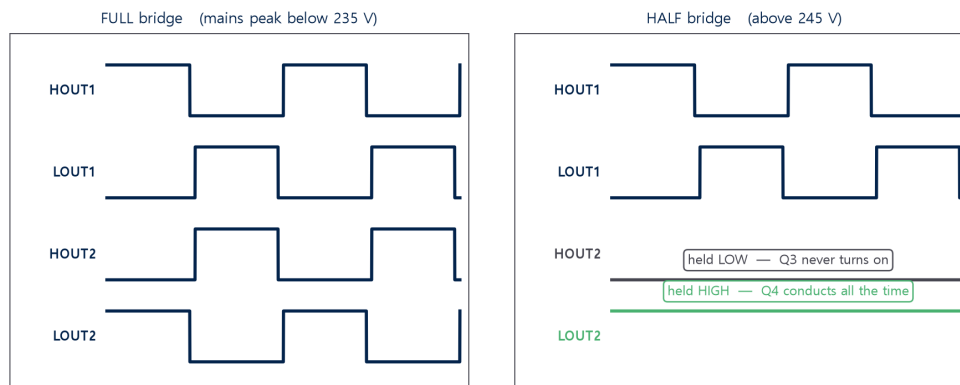
Q3 never turns on and Q4 never turns off so the leg-2 midpoint is ground. The bridge output now swings 0 to V_{in} ; C_r blocks the $V_{in}/2$ of DC, leaving the tank a $\pm V_{in}/2$ square wave. No extra hardware - the green device is simply held on.

Figure 25: Half bridge: leg 2 stops switching, and C_r removes the dc that the asymmetric drive creates.

The standing device is the hottest one. Q4 never switches, but it conducts the full tank current continuously — 10.9 A rms in this design, 9.65 W in one package. It is easy to overlook precisely because it is not switching, and it is the device that sets the heatsinking requirement.

5.3 How the controller does it

Only one gate signal changes between the two modes, which is why morphing costs nothing in hardware. The thresholds themselves are **fixed constants inside the IC** and cannot be moved by the designer; R_{CFG} decides only whether morphing is enabled at all and where the brown-out threshold sits.



The dead time between the two signals of a leg is what ZVS lives in. Morphing changes nothing else: same pins, same tank, same control law — leg 2 simply stops switching, and the transition is hysteretic so it cannot chatter.

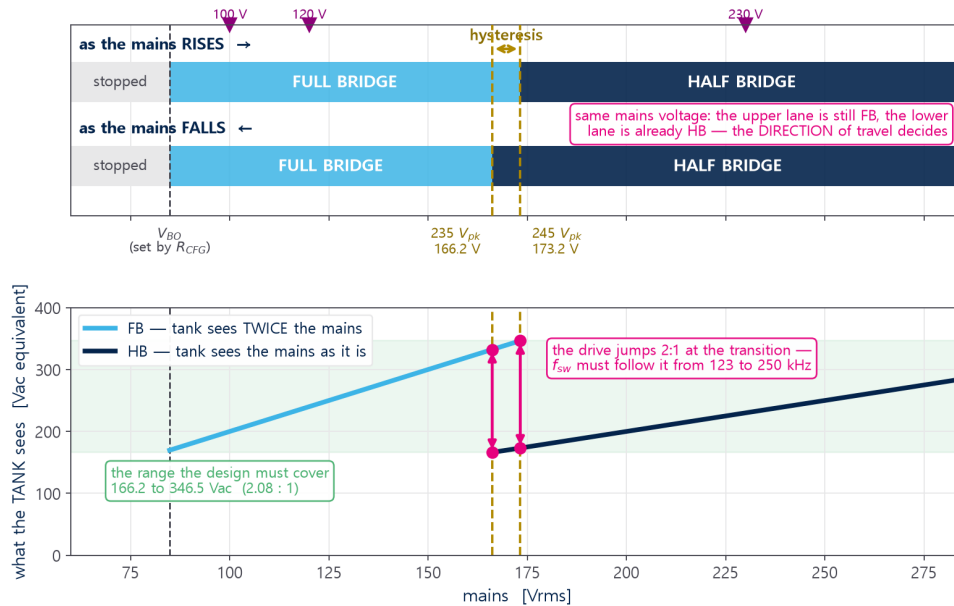
Figure 26: The four bridge drive signals in each mode. LOUT2 held high is the entire mechanism.

5.4 The hysteresis band, and the range it really implies

The two thresholds are not the same number: the part drops to half bridge at $245 V_{pk}$ going up and returns to full bridge at $235 V_{pk}$ coming down. Ten volts of hysteresis is what stops it chattering at the threshold, but it also means that **between 235 and $245 V_{pk}$ the mains voltage alone does not determine the mode** — the direction of travel does.

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Assigning each threshold to the side on which its mode is guaranteed gives the design range 173.2 to 332.3 Vac (1.92:1). Including the band gives **166.2 to 346.5 Vac (2.08:1)**. This design passes every check at the wider range as well, but a design with less margin would not, and the narrower figure must not be used when tightening a specification.



The 235 / 245 V_{pk} thresholds are FIXED IC constants. R_{CFG} decides only whether morphing is enabled and where V_{BO} sits. Real mains (purple markers) lies outside the band, so the mode is settled at start-up and stays.

Figure 27: Where each mode applies. Inside the band the mode depends on which direction the mains arrived from.

Where you will actually meet the band. Real mains never sits at 166 to 173 Vrms, so in service the mode is settled at start-up and stays. The band is met on a programmable ac source and in line dip and surge testing. Test it with a **step** across the threshold, not a ramp — a ramp gives the voltage loop time to follow, and a step does not.

Transition transient. Crossing a threshold changes the tank drive 2:1 within one line cycle, and f_{sw} must follow from 127 to 249 kHz against a voltage loop that crosses at 17.6 Hz. The output droops until the loop recovers. This event is **enveloped by the hold-up case**, which removes power entirely for 12 ms and is already met, so it is not a reason to enlarge C_{out} or speed the loop. That containment has not yet been confirmed in the time domain.

6 Design procedure

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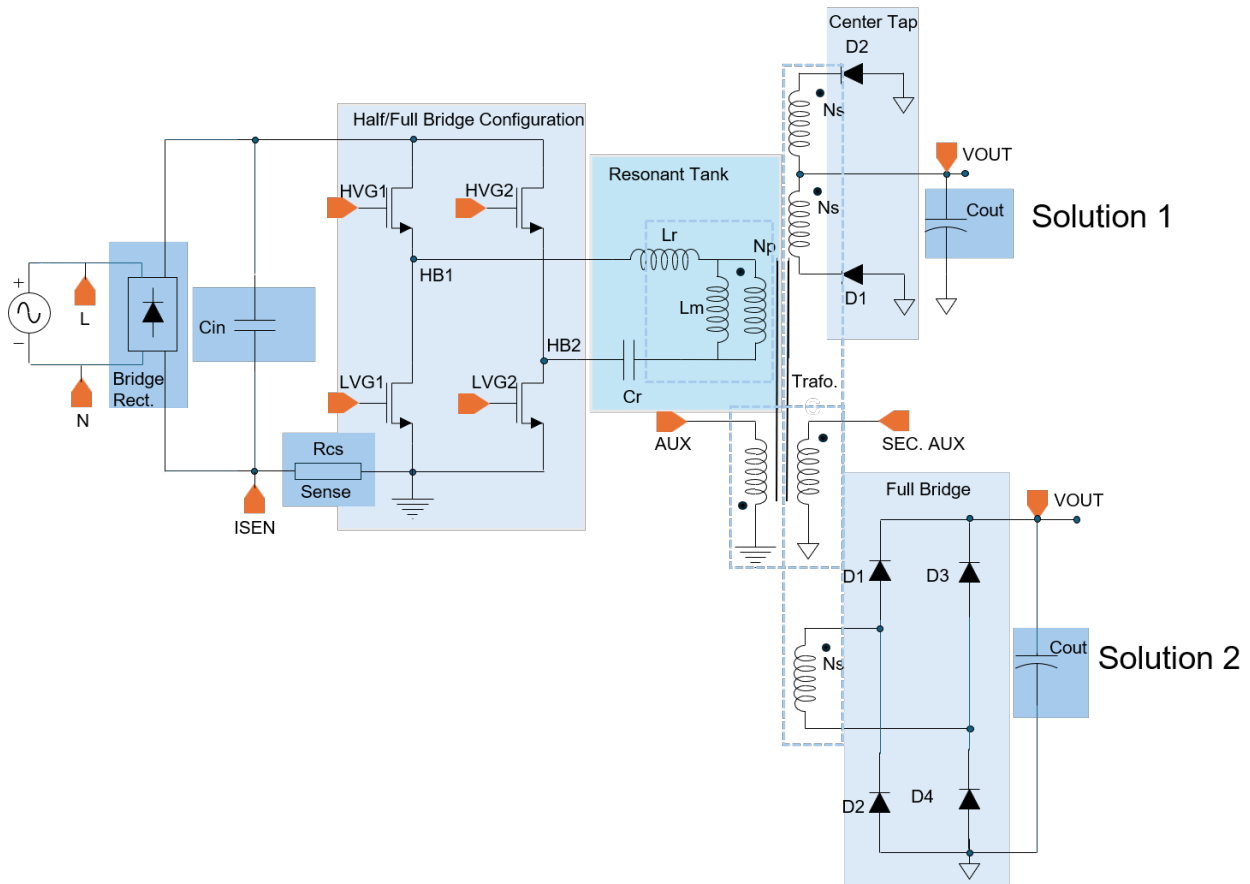


Figure 28: The complete power stage. The bridge rectifier feeds the half/full-bridge leg pair directly — there is no bulk capacitor and no boost stage. The secondary is drawn both ways: centre tap (Solution 1) and full bridge (Solution 2).

Pick the secondary before the tank: it sets $N_{rect'}$ and N_{rect} sets the reflected voltage. A centre tap puts one device in the conduction path, a full bridge puts two, so at a low output voltage the centre tap halves the secondary conduction loss. The price is twice the reverse voltage on each device, and a secondary winding that works only half of the time. This design uses the centre tap.

The sequence below is the order in which the quantities actually depend on one another. Working out of order will produce a tank that has to be redone.

6.1 The specification

The worked example in Section 7.1 states its specification in full, sorted by what kind of number each one is — imposed, chosen, assumed, or derived — because the only useful question about a number is whether it can be moved, and the answer differs by group. The procedure below refers to it but does not repeat it. Two of its qualifiers matter to every step here and are worth stating once. **The worst line frequency is the lowest**, so everything that depends on f_l uses 47 Hz. And **hold-up is specified at the worst line phase**, which is the ripple trough rather than the nominal output.

6.2 The equivalent input range

This is the step most often got wrong. With morphing the tank does not see 90 Vac at the bottom and 264 Vac at the top. It sees an **equivalent** input given by the bridge mode:

$$V_{ac,eq} = 2 V_{ac} \quad (\text{full bridge}) \quad V_{ac,eq} = V_{ac} \quad (\text{half bridge}) \quad (15)$$

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At the half-bridge edge the mains is $245 V_{pk}$ and the tank sees 173.2 Vac equivalent; at the full-bridge edge the mains is $235 V_{pk}$ in full bridge and the tank sees 332.3 Vac equivalent. **These two edges are the design corners, not the ends of the mains range:** the low edge is the worst case for gain and ZVS, the high edge for switching frequency.

6.3 Turns ratio and reflected voltage

The turns ratio only ever appears multiplied by the effective output voltage, so what matters is the **reflected voltage**:

$$V_{refl} = n V_{o, eff} = n(V_{out} + N_{rect} V_f) \quad (16)$$

Two ratios have to be kept apart. n is the equivalent-model ratio used in the gain equation; n_T is the physical turns ratio of a transformer whose leakage is integrated into L_r . Confusing them shifts the whole gain curve, and at $\lambda \approx 0.5$ they differ by more than 20 %. The transformer drawing must therefore specify open- and short-circuit inductance as well as turns.

6.4 The resonant tank

The equivalent ac load resistance for a centre-tapped secondary is

$$R_{ac} = \frac{4}{\pi^2} \frac{n^2 V_{o, eff}^2}{P_{in, LLC}} \quad (17)$$

with $P_{in, LLC}$ the power into the resonant stage. The 4 is the usual 8 with the *line-peak* power $2P$ in the denominator rather than the average — see the note below.

The gain requirement at the low equivalent corner and the ZVS requirement together cap the quality factor at Q_{ZVS} , and $R_{ac} Q_{ZVS}$ is then the impedance the tank is *sized* from: it fixes C_r through equation 1, and C_r with the target f_r fixes L_r . Three rules govern what happens next, and they matter more than the arithmetic:

- The three *calculated* figures are not one tank. C_r comes from the design impedance; L_r is then whatever pairs with the **selected** C_r , which is why it is not $R_{ac} Q_{ZVS} / 2\pi f_r$; and the L_m figure is L_r divided by the λ that *no load at the high corner* asks for — a condition a design may knowingly not meet (Section 4.8), so it is not a value to round towards.
- **Never round L_m up.** That lowers λ and takes the ZVS margin with it. C_r above its calculated value and L_m below have the opposite effect: Q_{pk} falls and λ is held.
- **Sit under the Q_{ZVS} cap rather than on it.** That gap is where most of a design's ZVS margin comes from, and it is a choice rather than an accident of rounding. L_m is chosen together with n so that $n_T = n\sqrt{1+\lambda_{act}}$ lands on a ratio that can actually be wound.

The same expression, evaluated somewhere else. Every LLC text writes $R_{ac} = (8/\pi^2)n^2 V_o^2 / P$, and so does this one. A two-stage converter has only one output power, so that is the end of it. Here the drawn power is $p(\theta) = 2P \sin^2 \theta$, so R_{ac} varies over the line cycle as well, and a single number has to be pinned to a chosen instant. Pin it to the **line peak**, where $p = 2P$ and the tank is most heavily loaded — putting $2P$ in the denominator of the same expression is what turns the 8 into a 4. **It is not a different formula and it has nothing to do with half bridge against full bridge;** the bridge factor lives in the equivalent input voltage, not in the load. Q then means the quality factor at the line peak, and every other phase scales from it as $Q(\theta) = Q_{pk} \sin^2 \theta$. Taking the two-stage number across instead understates the loading by two, and every tank value that follows is wrong.

6.5 ZVS verification

The closed-form ZVS estimate found in most design guides is a fitted approximation, and it is evaluated at the design Q_{ZVS} rather than at the Q the converter actually runs at. Measured against a full sweep it can err in **either**

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direction, by tens of per cent, and the sign of the error is not fixed — it depends on the tank. Optimistic is the dangerous direction, because it reports ZVS margin that is not there. Section 7.5 puts numbers on both directions.

And it has to be told which λ to use. The shortcut is written with a bare λ , and it means the *design* λ — the largest of the four candidates — together with the design Q_{ZVS} . Read the same expression with λ_{act} , which is what the symbol means everywhere after the tank is chosen, and the phase can come out **negative** — a capacitive answer for a tank that is well inside the inductive region. The sweep carries no such ambiguity: it uses λ_{act} and the Q of the moment, which is what the hardware has.

The reliable procedure is to sweep θ , input voltage and load with the **selected** tank and take the minimum. That is a grid, not a formula, and it is the only form of this check that can be trusted. Section 7.5 runs it for this design and says where the worst point turned out to be — which is not where a reader would guess.

6.6 Which side of resonance the converter runs on

Below f_r the secondary current is a truncated sine with a dead interval and the conduction ratio $d = f_{sw}/f_r$ is less than one; above f_r the current is continuous and d clamps at one. The rms figures that ratings and losses are built on carry d , so this is not a matter of wording.

Which side the converter lives on is not a separate choice. The turns ratio decides it. The demand is $M_{req} = 2nV_{o,eff}/(\sqrt{2}V_{eq})$, so a larger n asks for more gain, and more gain means further below resonance. Two candidate transformers on the same tank can end up on opposite sides of f_r at the same mains voltage.

Worth settling before the transformer is ordered. Comparing candidate turns ratios on gain margin alone hides it, and the two sides do not stress the secondary the same way: below resonance the rectifiers turn off at zero current, above it they do not.

6.7 Currents over the line cycle

Ratings come from the worst switching cycle; losses come from the line-cycle rms. They are different numbers and must not be confused. Two further rules apply to a single-stage converter in particular.

- The **composite** tank current is what the primary devices and the sense resistor see. It is the sum of the reflected load current and the magnetising current, and its peak is neither the sum nor the larger of the two peaks, because they occur at different instants.
- Below resonance the secondary conducts for only $d = f_{sw}/f_r$ of each half period. Omitting that conduction ratio over-estimates the rms, and the loss by the square of that.

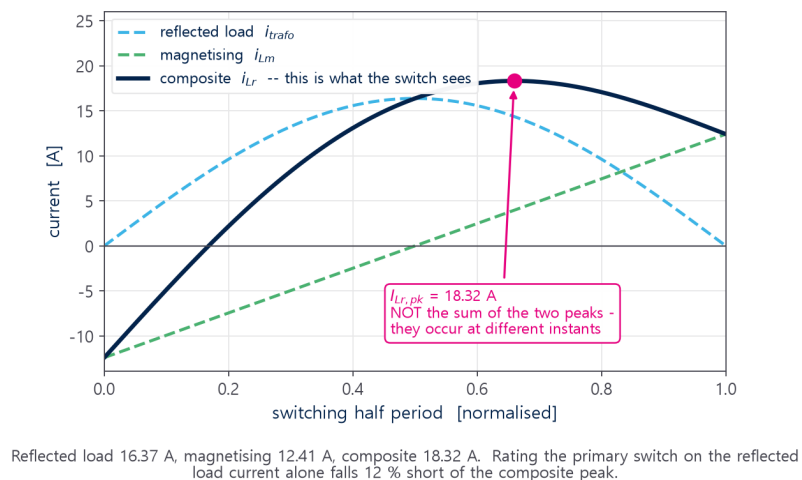


Figure 29: The composite tank current, and why its peak is not the sum of the two component peaks.

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6.8 The transformer

The tank has fixed L_r , L_m and a ratio; the transformer has turns, an open-circuit inductance and a leakage. Getting from one to the other is where most of the errors in a resonant design are made, because **two different turns ratios and two different inductances carry almost the same names**.

6.9 Two ratios, two inductances

When all of the leakage is referred to the primary, the tank sees an ideal transformer of ratio n , a series L_r and a shunt L_m . That n is **not** the wound ratio. With k the coupling coefficient,

$$n = k n_T, \quad k = \sqrt{\frac{L_m}{L_m + L_r}} = \frac{1}{\sqrt{1 + \lambda}} \quad (18)$$

so $n_T = n \sqrt{1 + \lambda}$. At $\lambda \approx 0.5$ the two differ by more than 20 %, which is far more than any gain margin.

The same split appears in the inductances. The tank model uses L_r and L_m ; the physical transformer is described by a magnetising inductance L_μ with leakage on both sides:

$$L_\mu = \sqrt{L_m (L_m + L_r)}, \quad L_{L1} = L_m + L_r - L_\mu, \quad L_{L2} = \frac{L_{L1}}{n_T^2} \quad (19)$$

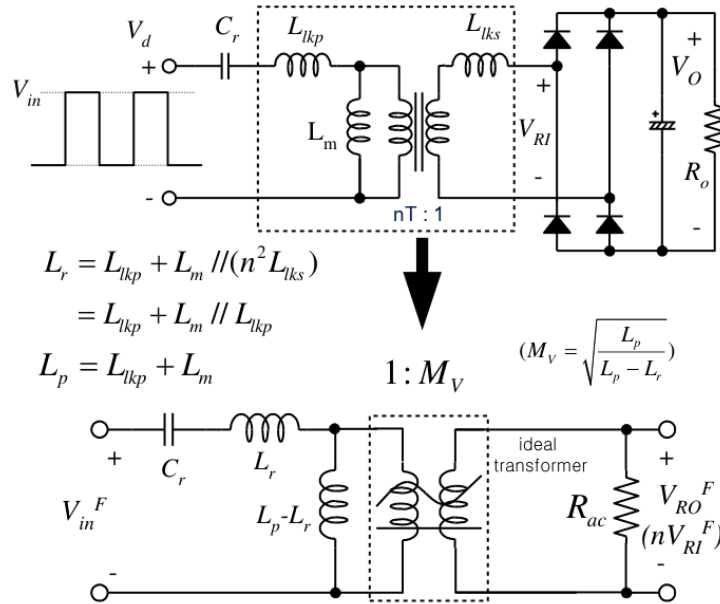


Figure 30: The same transformer drawn both ways. Above, the leakage split between the two sides; below, all of it referred to the primary so that the tank sees L_r , a shunt inductance and an ideal transformer whose ratio is no longer the wound one. The source calls the total primary inductance L_p , which is L_{open} here, and its M_v is $\sqrt{1 + \lambda}$ — so n_T/M_v is n . (ON Semiconductor AN-4151; the wound ratio relabelled n_T to match.)

That three-element form assumes the leakage splits evenly between the two sides. The real split is set by the winding arrangement rather than by the design, and it does not matter: the tank sees only the two quantities a meter can read.

6.10 What the transformer specification must say

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Only two inductances are measurable at terminals, and they are exactly the two the tank needs:

$$L_{open} = L_{\mu} + L_{L1} = L_m + L_r, \quad L_{short} = L_{L1} + \frac{L_{\mu} n_T^2 L_{L2}}{L_{\mu} + n_T^2 L_{L2}} = L_r \quad (20)$$

L_{open} is read at the primary with every other winding open, L_{short} with the secondary shorted. **Specify those two and the turns, never L_{μ}** — no terminal pair exposes L_{μ} , so a supplier cannot test it, and a part that meets it can still miss the tank. State the turns ratio in words as well: at one or two secondary turns the self-inductance of that winding is dominated by the loop of the lead-out, so $\sqrt{L_1/L_2}$ does not recover the ratio.

6.11 Peak flux is set by the secondary, not the primary

Below resonance the secondary conducts for a resonant half period and the output voltage stands across the secondary winding for all of it. Above resonance it conducts for a shorter switching half period, so f_r is the worst case everywhere on the line cycle:

$$B_{pk} = \frac{V_{out, eff}}{4f_r N_s A_e} \implies A_e \geq \frac{V_{out, eff}}{4f_r N_s B_{max}} \quad (21)$$

N_p does not appear. Adding primary turns does not reduce the flux; only adding secondary turns does. That is an uncomfortable result for a low-voltage, high-current output, where the secondary wants to be a single turn: halving N_s doubles the core area needed. That is why a single-turn secondary is expensive in core area even though it is the obvious choice for the current.

Computing the same flux from an inductance is correct only with the **physical** L_{μ} : $B_{pk} = L_{\mu} i_{\mu, pk} / (N_p A_e)$ reduces exactly to the expression above. Using the tank model L_m with the real N_p understates the flux by $\sqrt{1+\lambda}$ — about a quarter at $\lambda \approx 0.5$ — because the L_m branch carries $n V_{out}$ while the winding reflects $n_T V_{out}$.

6.12 The saturation test is not the peak winding current

A DC-overlap test leaves every other winding open, so none of the primary ampere-turns is cancelled by the secondary and the same current makes far more flux than it does in operation. Ask for the current that reproduces the operating flux in that test:

$$I_{eq} = \frac{B_{pk} N_p A_e}{L_{open}} \quad (22)$$

Handing the supplier the peak tank current instead asks for a flux density no ferrite reaches, and the answer comes back as an oversized core or as a query.

6.13 The specification, and what it is not allowed to leave out

The open-circuit tolerance is not a round number, and it is not free to be one. f_o goes as $1/\sqrt{L_{open}}$, so a low part raises the frequency floor towards the oscillator clamp. The fall a tank can tolerate follows from k_{floor} and is typically a few per cent, so the reflex $\pm 10\%$ can let a compliant part hard switch. Section 6.18 derives the figure and gives the two ways out. Ask the supplier for the asymmetric limit, or buy the room in the oscillator first and then ask for $\pm 10\%$.

Core, bobbin, wire and winding order are the supplier choice, and insulation and creepage follow the applicable safety standard. Nothing above constrains them; everything above is measurable at the terminals.

6.14 If one transformer is not practical

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At a low output voltage and high current the secondary is a single heavy turn and the core comes out large, which can lose against a height or footprint limit. The transformer can then be built as N_x identical units with **primaries in series and secondaries in parallel**. Series primaries carry the same current, so the paralleled secondaries share automatically without balancing resistors.

- Open-circuit and leakage inductance divide by N_x , and so do the secondary currents. **The primary current does not divide.**
- The DC-overlap test current does not change: flux and primary ampere-turns scale together.
- The wound ratio of the assembly becomes $N_x N_p / N_s$, so **the achievable ratios are quantised in steps of N_x / N_s** . That has to be respected when the tank is chosen, not discovered afterwards.

6.15 The input capacitor

With no bulk capacitor, C_{in} is a film capacitor that absorbs switching ripple and nothing else. Making it large is not conservative: it holds charge across the zero crossing and distorts the line current, so it is a THD term, not a filter term.

$$C_{in} = 3 \frac{nF}{W} \times P_{in} \quad (23)$$

That is a **lower** bound, so round up to the next standard value and rate it for the full line voltage.

6.16 The output capacitor bank

Two conditions apply, and the larger wins. The ripple condition is

$$C_{out} \geq \frac{P_{out}}{2\pi f_{l,min} \Delta v V_{out}^2} \quad (24)$$

and the hold-up condition is

$$C_{out} \geq \frac{2P_{out}T_{hold}}{(V_{out} - \frac{1}{2}\Delta v_{pp})^2 - V_{o,min}^2} \quad (25)$$

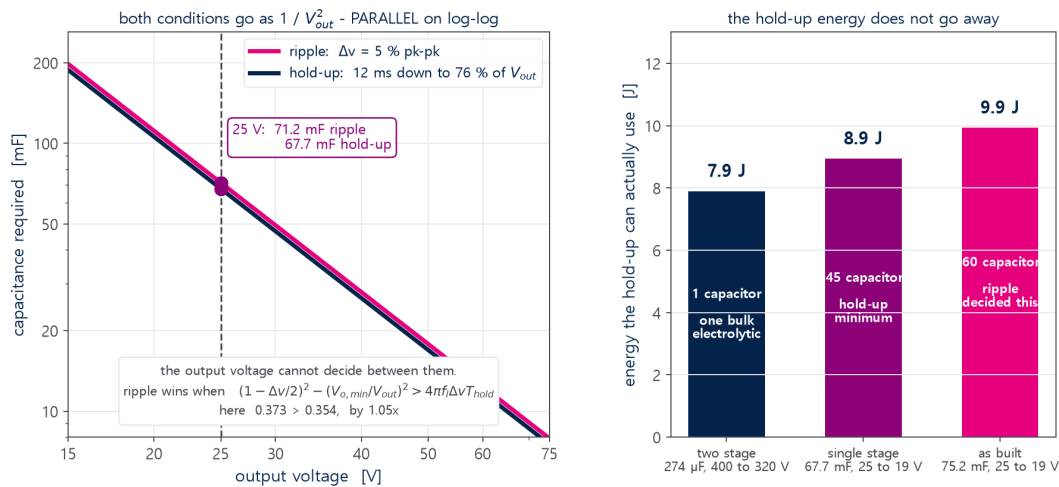
Hold-up starts at the worst line phase, and that belongs in the sizing. If the mains disappears at the trough of the $2f_l$ ripple the bank is already half a ripple below nominal when it starts, which is why V_{out} above carries the $-\frac{1}{2}\Delta v$ term. The difference is not small: it costs about a tenth of the ride-out time. Note also that Δv_{pp} there is the ripple the bank actually produces, not the allowance. The condition therefore depends on its own answer, which is why it is checked once the bank is chosen.

Which one wins is a question about the specification and not about the output voltage: both conditions scale as $1/V_{out}^2$, so V_{out} cancels when they are divided. Writing $k = V_{o,min}/V_{out}$, ripple dominates when

$$\left(1 - \frac{\Delta v}{2}\right)^2 - k^2 > 4\pi f_l \Delta v T_{hold} \quad (26)$$

This screening form uses the *allowed* Δv rather than the achieved ripple, because it is asked before the bank exists. On the specification below the two come out close, and loosening Δv to 10 % would hand the decision to hold-up instead.

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Removing the boost stage removes the 400 V bus, not the energy it held. The first two bars are the SAME 7.9 J - 12 ms at full power - and the capacitance between them differs by $(400^2 - 320^2) / ((25 - 0.6)^2 - 19^2) = 245\times$. That single line is the cost of the architecture. The bank actually built is larger again because RIPPLE, not hold-up, set its size.

Figure 31: Both sizing conditions fall as $1/V_{out}^2$, so only the specification separates them. Right: hold-up asks for the same energy either side of the boost stage — what changes is the capacitance that holds it.

- Include the $2f_s$ component in the ripple current. The switching-frequency part alone is well short of the truth: the $2f_s$ envelope carries a comparable rms of its own, and the two are orthogonal so they add in quadrature.
- With a centre-tapped secondary, judge the capacitor ripple current at the **output node**, not per winding.
- Use the ESR at the **switching frequency**. An electrolytic $\tan\delta$ is quoted at 120 Hz and is five to ten times larger there.

A risk this architecture creates. The controller start-up window is finite, and a bank of this size is a far heavier start-up load than the few millifarads a conventional design presents. Cold start into the full bank should be one of the first things measured on hardware.

6.17 Semiconductor requirements

State requirements rather than pick parts: what to pick depends on package, thermal design and cost. Four rules decide whether the requirement is even stated correctly.

- Rate the primary switches on the **composite tank peak**, not on the reflected load current. The load component alone under-rates the device by roughly a tenth, because the magnetising current is missing from it.
- Divide currents by the number of devices actually in parallel before judging stress. Tabulated currents are **per switch position** on the primary and **per leg** on the secondary; a centre-tapped leg carries the whole secondary current, not half of it.
- Compute loss from $R_{DS(on)}$ at $T_{j,max}$, not at 25 °C. The datasheet maximum is process spread at 25 °C; temperature is a separate multiplier, roughly 1.8 for a 600 V superjunction device and 1.9 for a low-voltage synchronous rectifier. Using the 25 °C value to size a heatsink is optimistic by about a factor of two.
- A curve plotted against T_a may still be read as T_j if it is a **pulse test** — the die does not self-heat during the pulse. For any non-pulsed curve this is not valid.

The standing device in half-bridge morphing is the one to watch. It never switches, so it is easy to overlook, and it carries the whole tank current continuously. On a design of this power it is where the conduction-loss budget is either met or knowingly missed, and Section 7.10 shows which of the two happened here.

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6.18 Controller network

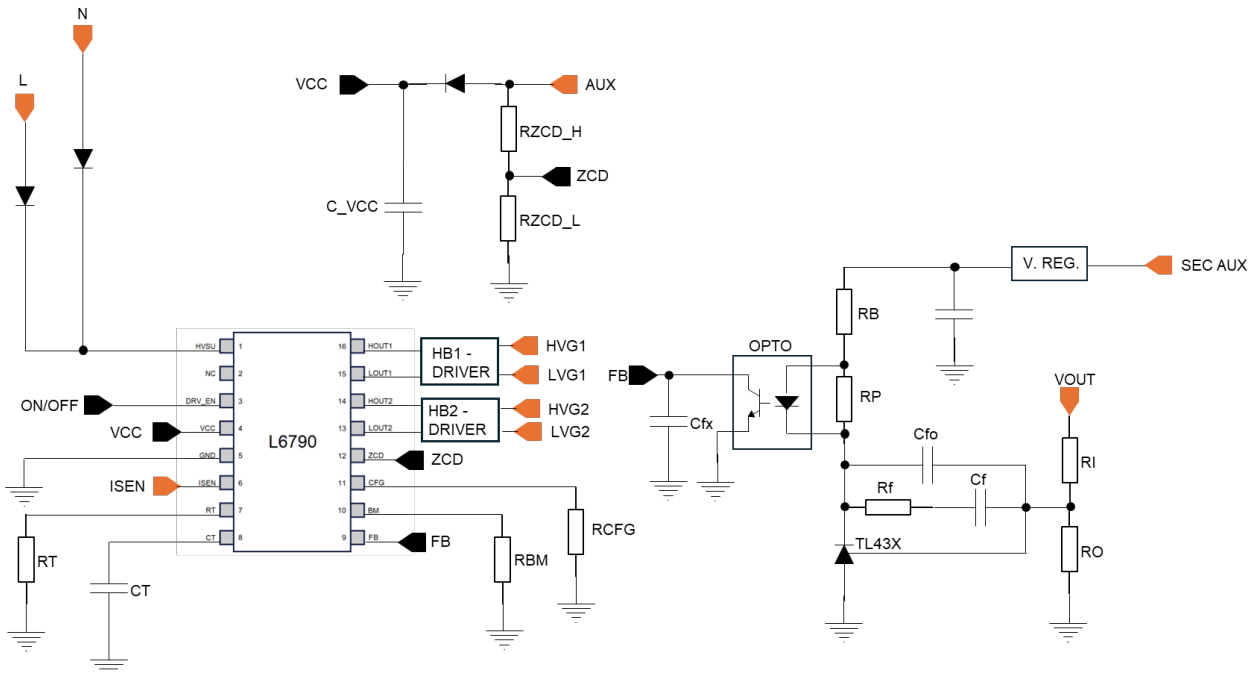


Figure 32: The controller and its passive network. HVSU is fed from the AC side of the bridge; the auxiliary winding drives ZCD through a divider; R_T and C_T set the oscillator limits; R_{CFG} and R_{BM} are read at power-up.

Decide first what the auxiliary winding is for, because it changes one of the rules below. In the usual arrangement it does two jobs — it senses the zero crossing and it supplies V_{CC} — and the second job caps its turns ratio, since the rectified auxiliary voltage at OVP2 must stay under the V_{CC} rating. That cap is often the binding constraint on the winding. Where V_{CC} is supplied from elsewhere, as it is here, the cap does not exist: the winding senses only, its voltage is set by whatever ratio the ZCD divider is designed around, and the sole remaining constraint is that the turns come out whole. A ratio check carried over from the self-supplied case will read as a failure in that arrangement and should be marked not applicable rather than satisfied.

Each of these parts follows from something already fixed, and Section 7.13 sizes them one at a time in the order they have to be done. The one that deserves the most attention throughout is R_T , because it sets the oscillator floor and that floor must stay above f_0 .

That floor is usually the thinnest margin in the design. f_{Min} must sit above f_o , or nothing but the anti-capacitive protection stands between the converter and the capacitive region. It also depends on the oscillator idle time, which the draft datasheet states inconsistently, so whatever it computes to has to be confirmed by measuring $f_{\text{sw}}(\theta)$ on hardware.

That ratio is also what the transformer tolerance has to fit inside, and this is easy to miss because the two live in different documents. Since $f_o = 1/(2\pi\sqrt{L_{open}C_r})$, a **low** open-circuit inductance raises f_o . The fall that can be tolerated is

$$\frac{\Delta L}{L} = 1 - \frac{1}{k_{floor}^2} \quad (27)$$

A floor margin of a few per cent allows a fall of a few per cent in inductance. A transformer drawing asking for the usual $\pm 10\%$ does **not** fit inside that: at the bottom of the tolerance the converter hard switches. There are two ways out — tighten the tolerance on the low side, or lower R_T to lift f_{Min} and make the room. **Check this before the transformer is ordered**, not after.

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6.19 Voltage loop and compensation

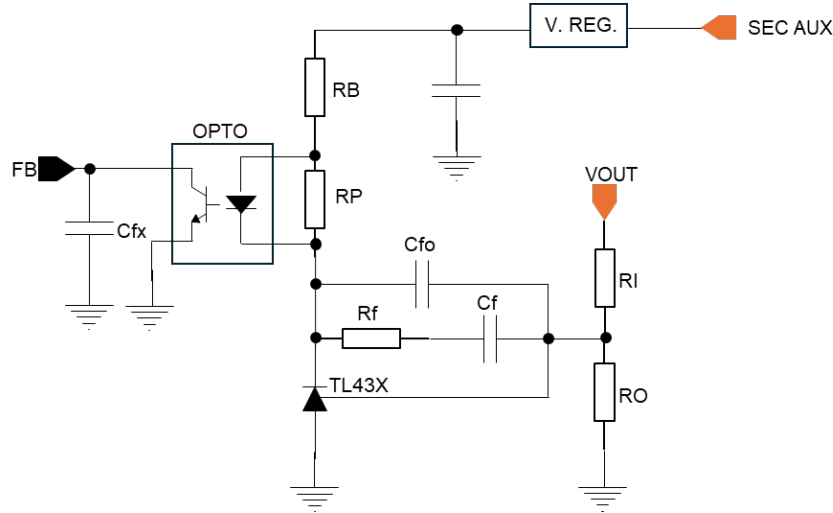


Figure 33: The secondary-side error amplifier and the optocoupler. R_I and R_O set the regulated output, $R_f C_f$ and C_{fo} shape the compensator, and R_p and R_B bias the shunt regulator and the optocoupler diode.

The output is a capacitor fed by a constant-power source, so the control-to-output transfer is a **pure integrator**: -90° at every frequency. That makes the loop easy to compensate and hard to place, because the crossover is set by a constraint that has nothing to do with stability.

Any $2f_i$ ripple that survives the loop and reaches the feedback pin becomes a power command that varies over the line cycle, and therefore input current distortion — mostly third harmonic. Raising the crossover reduces output ripple and **increases** distortion. The two constraints pull in opposite directions, and the practical answer is in the high teens of hertz.

The compensator is a Type II network placed by the Venable K-factor method. With α_v the ratio between the error-amplifier gain at $2f_i$ and the gain the loop needs there,

$$K_v = \frac{1}{2\alpha_v} \left[(1 + \alpha_v^2) \tan \Phi_M + \sqrt{(1 + \alpha_v^2 \tan \Phi_M)^2 + 4\alpha_v^2} \right] \quad (28)$$

K_v puts the zero and the pole the same factor either side of the crossover. That is what lands the phase bump where it is needed:

$$f_{MB} = \frac{1}{2\pi} \sqrt{\frac{G_o K_v E A_o}{\Gamma_v}}, \quad f_p = K_v f_{MB}, \quad f_z = \frac{f_{MB}}{K_v} \quad (29)$$

f_{MB} is the point they sit around: the zero goes below it and the pole above it, by the same factor. A third pole f_{px} is added near a kilohertz to roll off switching noise.

6.20 Gain margin, and why it needs checking

Phase margin alone does not close the stability argument. Above f_{px} this network has one zero against two poles, so the phase heads for -270° and -180° is **crossed at a finite frequency**. That crossing needs no root search either: setting the zero and pole phase contributions equal and taking the tangent of both sides leaves one square root,

$$f_{180} = \sqrt{f_p f_{px} - f_z (f_p + f_{px})}, \quad GM = -20 \log_{10} |T(f_{180})| \quad (30)$$

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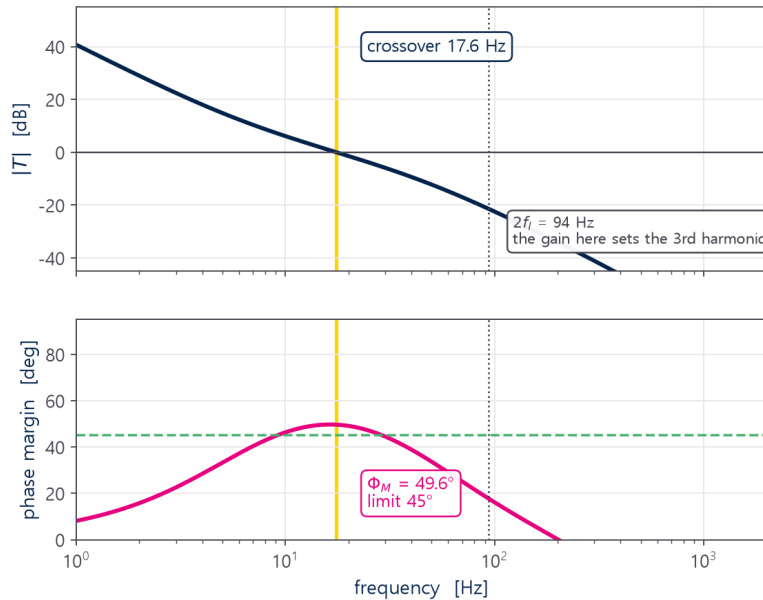
A negative value under the root is the answer, not an error. It means the phase never reaches -180° , so the gain margin is infinite. Judge the result against 6 dB as a floor and 10 dB as a comfortable target. With a crossover in the tens of hertz the margin is usually large in this topology — which is a reason to compute it, not a reason to assume it.

6.21 Feedback ripple against the burst threshold

A crossover this low leaves $2f_l$ ripple on the feedback pin. Because that pin is a power command, the ripple walks the commanded power up and down every half line cycle; if it straddles the burst-entry threshold the converter chatters in and out of burst. The ripple amplitude at the burst point is

$$\Delta V_{FB} = \frac{P_{in,BM}}{V_{out}} \frac{1}{2\pi f_l C_{out}} G_{EA}(2f_l) \quad (31)$$

Lowering R_{BM} by $\Delta R_{BM} = \Delta V_{FB} / (2 \times 0.01 \text{ V/k}\Omega)$ moves the burst entry point below that ripple. **Raising the crossover would shrink the ripple but worsen the distortion**, so trading it against R_{BM} is the practical answer rather than speeding up the loop.



Open-loop gain of the selected compensation network. Raising the crossover makes the loop chase the output ripple and puts third harmonic on the input current.

Figure 34: Open-loop gain and phase margin of the selected compensation network.

A loop this slow cannot recover from a large load step. The controller's anti-saturation circuit does that instead, which is why the optocoupler bias resistor must sit inside its permitted window. Checking the window is not optional.

7 Design example

Everything so far is method: what each quantity is, which equation gives it, and what goes wrong if it is worked out the wrong way.

Here are the numbers. One design runs from the specification down to component values. Each value names the equation it came from and shows the inputs put into it, so the arithmetic can be checked. The specification is **90 to 264 Vac in, 25 V / 26.3 A = 657.5 W out**, centre-tapped synchronous rectification, no bulk capacitor and no boost stage.

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7.1 The specification, sorted by what kind of number it is

A specification sheet usually runs all of its numbers together in one table, and that is where a good deal of later trouble starts. Four different kinds of quantity are in there. Some are imposed from outside and cannot be traded. Some were chosen by the designer and could have been chosen otherwise. Some are assumptions standing in for a measurement nobody has made yet. And some are simply what the first three produce. They are kept apart here, because the only useful question about a number — can I move it? — has a different answer in each group.

Item	Symbol	Value
Mains input	V_{ac}, f_l	90 to 264 Vac, 47 to 63 Hz
Output	V_{out}, I_{out}	25 V, 26.3 A (657.5 W)
Output ripple, $2f_l$ pk-pk	Δv	$\leq 5\%$ of V_{out}
Hold-up	$T_{hold}, V_{o,min}$	12 ms down to 19 V, at 100 Vac, worst line phase

Table 2: Set by the load and the mains. Not the designer's to change.

Two of these four carry a qualifier that is easy to drop and expensive to drop. **The worst line frequency is the lowest**, because the longest line period gives the largest ripple and the shortest hold-up, so everything that depends on f_l uses 47 Hz here, not 50 or 60. And **hold-up is specified at the worst line phase**, which means the bank starts half a ripple below nominal rather than at V_{out} — worth 11 % of the ride-out time here, as Section 7.9 shows.

Item	Symbol	Value	Consequence
Secondary rectifier	N_{rect}	1 (centre tap, SR)	halves the secondary conduction loss; doubles the device reverse voltage
Target series resonance	f_r	150 kHz	sets the magnetics size and, with C_r , fixes L_r
Specified maximum f_{sw}	$f_{sw,max}$	225 kHz	a $1.5 \times f_r$ convention, and it is what sets λ — Section 4.7
Bridge dead time	t_D	220 ns	the time the ZVS sweep has to beat
V_{CC} source	—	external 12 V rail	removes the turns-ratio ceiling on the auxiliary winding

Table 3: Chosen here. Each could have been chosen differently; the section named is where that shows up.

Item	Symbol	Assumed	If it is wrong
LLC stage efficiency	η_{HB}	98 %	R_{CS} and R_{ac} move by about 3 % for a 95 % outcome — nothing downstream is sensitive
Rectifier forward drop	$N_{rect} V_f$	0 V	a synchronous rectifier drops $IR_{DS(on)}$, carried in the loss budget rather than in the reflected voltage
Oscillator idle time	T_{idle}	250 ns	the draft datasheet also says 700 ns, which would close the floor margin entirely — the first thing to measure
$R_{DS(on)}$ temperature factor	k_T	1.8 / 1.9	primary / secondary, read off datasheet curves at $T_{j,max}$; a wrong factor mis-sizes the heatsink directly
Burst entry point	r_{BM}	75 W (11 % of rated)	sets R_{BM} , which then has to be checked against the feedback ripple

Table 4: Assumed, in place of a measurement. Replace each one from the first prototype.

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Item	Symbol	Value	From
Output ripple in volts	Δv_{pp}	1.18 V	5 % of 25 V
Reflected output voltage	$V_{o,eff}$	25.0 V	$V_{out} + N_{rect} V_r$, with $V_r = 0$
Rated input power	P_{in}	684.8 W	P_{out} through the bridge, EMI and LLC loss budgets
Overall efficiency	η_{tot}	96.02 %	P_{out}/P_{in}
Equivalent input range	$V_{ac,eq}$	173.2 to 332.3 Vac	the morphing edges, not the mains range — next section

Table 5: Follows from the three tables above — arithmetic, not input.

7.2 What the design came out as

Block	Values
Tank	C_r 100 nF, L_r 11 μ H, L_m 20 μ H, λ 0.55, f_r 151.7 kHz, f_o 90.4 kHz
Transformer	$N_p:N_s$ 15:2, n 6.02, n_T 7.50, L_{open} 31.0 μ H, L_{short} 11.0 μ H, $A_e \geq 103$ mm ² at B_{pk} 0.20 T
Frequency	f_{sw} 127.1 kHz at the low corner, 249.0 kHz at the high corner
Output	470 μ F $\times$ 160 = 75.2 mF, ripple 1.18 V (4.74 %), hold-up 13.42 ms
Controller	R_T 11 k Ω , C_T 470 pF, R_{CS} 24.0 m Ω , R_{CFG} 30 k Ω , R_{BM} 30 k Ω
Loop	f_{cross} 17.57 Hz, Φ_M 49.64°, third harmonic 4.23 %

Table 6: Principal values.

7.3 What a verification margin is

Every check has the same shape. Take what the design gets, X_{act} , divide it by what the design needs, X_{req} , and call the answer k :

$$k = \frac{X_{act}}{X_{req}} \implies \text{pass when } k > 1 \quad (32)$$

That is the whole definition. $k = 1.05$ is five per cent of room, $k = 0.9$ is ten per cent short, and the number is read directly as a factor rather than as a difference.

Why force every check into the same shape? Because the raw quantities do not agree on which way is good. T_{ZC} should be *large*. f_o should be *small*. A loss should be small, a hold-up time long. As raw numbers, a reviewer has to remember the direction of each one.

As ratios they all point the same way, and they have no units. So a frequency check and a power check can sit in the same column and be compared. The smallest k in the table is the thinnest part of the design, whatever it happens to be about.

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Check	k = has / needs	Substituted	k
ZVS at the worst point of the sweep	$T_{ZC,min} / t_D$	401 ns / 220 ns	1.824
Oscillator floor clears the lower resonance	f_{Min} / f_o	92.25 kHz / 90.39 kHz	1.021
Oscillator ceiling clears the operating maximum	$f_{Max} / f_{sw,max}$	319.61 kHz / 249.03 kHz	1.283
Over-current threshold clears the tank peak	$I_{OCP1} / I_{Lr,pk}$	22.92 A / 18.32 A	1.251
Hold-up achieved against hold-up required	t_{hold} / T_{hold}	13.42 ms / 12 ms	1.119
Secondary loss against its budget	$P_{budget} / P_{SR,leg}$	3.00 W / 2.83 W	1.063
Primary loss against its budget	$P_{budget} / P_{mos,dc}$	3.00 W / 9.65 W	0.311

Table 7: The verification margins, and the ratio each one is.

k below 1 is a result, not a failed build. $k_{Ploss} = 0.311$ says the standing primary device spends 9.65 W against a 3 W budget. The budget is itself a choice, so the ratio could be made to pass by writing 10 W into the numerator — at which point the check stops checking anything. It is left as it is, and Section 6.17 says why the design goes ahead anyway.

Do not read two of these as percentages. $k_{floor} = 1.021$ looks like a safe 2 %. But two things move it — the transformer open-circuit tolerance and the oscillator idle time — and neither is known to 2 % yet. k_{ZVS} is the same: its numerator came from a sweep, not a formula. The rest of the table is arithmetic on values that are already fixed.

7.4 Following the numbers through

The table above is the destination. This is the road: every quantity in the order it was actually computed, with what it was computed from. Read down the “from” column and it is a chain — nothing in it appears before the things it needs, which is the whole reason the design procedure is in that order.

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#	Quantity	Value	Eq.	From
1	Input power P_{in}	684.8 W	—	P_{out} 657.5 W plus the bridge, EMI and LLC loss budgets
2	Power into the tank $P_{in,LLC}$	670.9 W	—	P_{in} less the bridge and EMI budgets
3	Equivalent input, low corner	173.2 Vac	15	$245 V_{pk}$ in half bridge, $\div\sqrt{2}$
4	Equivalent input, high corner	332.3 Vac	15	$235 V_{pk}$ in full bridge, $\times 2 \div \sqrt{2}$
5	Turns ratio n	6.024	18	chosen with L_m so that n_T lands on a windable ratio — the plain calculation gives 6.046
6	Reflected voltage	150.6 V	16	$n V_{o,eff}$
7	Gain demanded at the low corner	1.2294	9	$2 n V_{o,eff} / (\sqrt{2} \times 173.2)$
8	Gain demanded at the high corner	0.6409	9	the same expression at 332.3 Vac
9	AC load resistance R_{ac}	13.70 Ω	17	$(4/\pi^2) n^2 V_{o,eff}^2 / P_{in,LLC}$ — the 4 is the line peak
10	Quality-factor cap Q_{ZVS}	1.7019	4	the smaller of the gain and ZVS limits at the low corner
11	Design impedance	23.32 Ω	—	$R_{ac} Q_{ZVS}$
12	C_r	45.50 $\rightarrow$ 100 nF	1	$1/(2\pi f_r \times \text{design impedance})$, then rounded up hard to buy Q margin
13	L_r	11.26 $\rightarrow$ 11 μ H	1	whatever pairs with the selected C_r at $f_r = 150$ kHz
14	λ required	1.241	13	the largest of four candidates — two of them carry $f_{sw,max}$ in a denominator
15	L_m	8.87 $\rightarrow$ 20 μ H	—	not that ratio: chosen with n , knowingly missing the no-load condition
16	Realised λ_{act}	0.550	—	L_r/L_m of the selected parts
17	Realised f_r, f_o	151.7 / 90.4 kHz	1, 2	from C_r, L_r and L_r+L_m
18	Realised Q_{pk}	0.766	3	$\sqrt{(L_r/C_r)} = 10.49 \Omega$, divided by R_{ac} — well under the cap
19	Physical turns ratio n_T	7.500	18	$n \sqrt{(1+\lambda_{act})}$, landing on 15:2

Table 8: The design, step by step.

Six of those rows do the real work, and they are worth seeing opened out. Each one is the equation named beside it, with this design's numbers put in:

- **Row 5, the turns ratio.** Equation 18 is inverted: $n = n_T / \sqrt{(1+\lambda_{act})} = 7.500 / \sqrt{(1+0.550)} = 7.500 / 1.2450 = \mathbf{6.024}$. The wound ratio is the input here, not the output, because 15:2 is what can be built.
- **Row 7, the gain the tank is asked for.** Equation 9 at $\theta = 90^\circ$: $M = 2nV_{o,eff} / (\sqrt{2}V_{ac,eq}) = 2 \times 6.024 \times 25.0 / (1.4142 \times 173.24) = 301.2 / 245.0 = \mathbf{1.2294}$.
- **Row 9, the load the tank sees.** Equation 17: $R_{ac} = (4/\pi^2) n^2 V_{o,eff}^2 / P_{in,LLC} = 0.4053 \times 36.29 \times 625 / 670.9 = \mathbf{13.70 \Omega}$. Note the 4, not the textbook 8: $P_{in,LLC}$ is the line-peak power.
- **Row 12, the resonant capacitor.** The design impedance and the target f_r give it through equation 1 rearranged: $C_r = 1/(2\pi f_r R_{ac} Q_{ZVS}) = 1/(2\pi \times 150 \text{ kHz} \times 23.32 \Omega) = 45.50 \text{ nF}$, taken to **100 nF**.
- **Row 13, the resonant inductor.** Same equation, now solved for L with the selected C_r : $L_r = 1/((2\pi f_r)^2 C_r) = 1/((2\pi \times 150 \text{ kHz})^2 \times 100 \text{ nF}) = 11.26 \mu\text{H}$, taken to **11 μ H**. This is why L_r is not $R_{ac} Q_{ZVS} / 2\pi f_r$; C_r moved first.

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- **Row 18, the quality factor the tank actually runs at.** $Z_0 = \sqrt{L_r/C_r} = \sqrt{(11 \mu\text{H} / 100 \text{ nF})} = 10.49 \Omega$, and $Q_{pk} = Z_0/R_{ac} = 10.49/13.70 = \mathbf{0.766}$, against the 1.702 cap of row 10.

Steps 12, 15 and 18 are where the judgement is. Everything else in that table is arithmetic that any two engineers would reproduce identically. C_r was taken to 100 nF against a calculated 45.5 nF and L_m to 20 μH against a calculated 8.9 μH , and the two together are why the tank runs at $Q_{pk} = 0.766$ instead of at the 1.702 cap. That gap is most of the ZVS margin this design has.

Load	173.2 Vac eq.	180.0 Vac eq.	225.0 Vac eq.	264.0 Vac eq.	332.3 Vac eq.
100 %	460	517	671	674	603
75 %	757	776	791	720	561
50 %	1084	1079	971	808	501
25 %	1470	1445	1231	974	401

Table 9: T_{ZC} [ns] over the operating space, selected tank. The dead time is 220 ns.

7.5 ZVS over the whole operating space

The closed form of Section 6.5 reads 292 ns on this tank against a swept 460 ns, so it understates the margin by 36 %; on another tank in the same family it overstates by about 23 %. Read with $\lambda_{act} = 0.550$ instead of the design $\lambda = 1.241$ it returns -471 ns — capacitive, for a tank that is well inside the inductive region. So the sweep is what this design is held to.

The worst point is not the corner the tank was designed at. Full load at the low equivalent input is where the gain requirement is set, and it gives $T_{ZC} = 460$ ns, a margin of 2.09. The smallest number in the table is **401 ns at 25 % load and the 332.3 Vac equivalent corner**, a margin of 1.82 against $t_D = 220$ ns. That is the number this design is held to.

Why light load at high line can be the ZVS corner. The charge that swings the bridge node is carried by the magnetising current, and its peak goes as $V_{in}/(4f_{sw}L_m)$. Shed load and the controller raises f_{sw} to cut the gain, so that current falls even though the node still has the same capacitance to move. Whether this beats the full-load corner depends on the tank: it does not on a high turns ratio, where the design already sits deep below resonance, and it does on a lower one. **Sweeping only the design corner will therefore report a margin that is not there**, which is the same trap as trusting the closed form.

7.6 Which side of resonance this design runs on

Section 6.6 says the turns ratio decides this. With $n_T = 7.50$ the answer is below. At all but the lowest line conditions the tank spends part of every line cycle above f_r . There the secondary loses zero-current turn-off, so the rectifier body diode and the SR dead time both need checking.

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Line condition	V_{eq}	peak f_{sw}	side of f_r
minimum line	180 V	131.3 kHz	below
tank minimum	173 V	127.1 kHz	below
110 Vac	220 V	156.4 kHz	above
nominal	225 V	159.7 kHz	above
230 Vac	230 V	163.2 kHz	above
maximum line	264 V	188.5 kHz	above
FB threshold	332 V	249.0 kHz	above

Table 10: Peak f_{sw} over the half cycle against $f_r = 151.7$ kHz. 5 of the seven line conditions cross into above-resonance operation for part of the cycle.

7.7 The currents this design has to carry

Ratings come from the worst switching cycle and losses from the line-cycle rms, so both are listed and they are different numbers. The row to watch is the composite tank peak: it is what the primary devices and R_{CS} actually see, and it is neither the sum of the two component peaks nor the larger of them.

Quantity	Value	Where it is used
f_{sw}	127.1 kHz	oscillator floor check
$I_{sec,pk}$	98.6 A	secondary rectifier peak
$I_{trafo,pk}$	16.37 A	reflected load component
$I_{Lm,pk}$	12.41 A	magnetising component
composite tank peak	18.32 A	primary device rating, OCP margin
$I_{pri,rms}$ (worst cycle)	13.76 A	primary rms rating
I_{pri} (line-cycle rms)	10.93 A	primary conduction loss
I_{rect} per leg (line-cycle rms)	28.34 A	secondary conduction loss
I_{Cout} (line-cycle rms)	30.24 A	output bank ripple current

Table 11: Currents at the low equivalent corner, full load, $\theta = 90^\circ$.

Quantity	Calculated	Selected	Direction
C_r	45.50 nF	100 nF	rounded up
L_r	11.26 μ H	11 μ H	rounded down
L_m	8.87 μ H	20 μ H	not a rounding — see below
$\lambda_{act} / f_r / f_o$	—	0.550 / 151.7 kHz / 90.4 kHz	from the selected parts

Table 12: Tank: calculated against selected.

7.8 The transformer, as built

The tank asks for $L_r = 11$ μ H, $L_m = 20$ μ H and $n_T = 7.50$. At 26.3 A out the secondary wants to be one or two heavy turns, and a single core that can carry the flux at that few turns comes out too large for the enclosure. This design therefore uses 3 identical units, primaries in series and secondaries in parallel, as Section 6.14 describes.

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Quantity	Per unit	Assembly	Note
Turns	5 : 2	15 : 2	primaries in series, secondaries in parallel
Open-circuit inductance	10.3 μ H	31.0 μ H	$L_r + L_m$, measured with every other winding open
Short-circuit inductance	3.67 μ H	11.0 μ H	this is L_r : no separate resonant inductor
A_L	413 nH	—	open-circuit inductance / N_p^2
Peak flux density	182 mT	—	$V_{o,eff}/(4 f_r N_s A_e)$ — N_p does not enter
Core area	$\geq 103 \text{ mm}^2$	—	to hold B_{pk} at 0.20 T; EE6405 offers 113 mm^2 , a factor of 1.098
Saturation test current	14.9 A	—	flux-equivalent 10.0 A rounded up, not the 18.3 A tank peak

Table 13: The transformer as it is actually wound and specified.

Three of those rows are computed rather than specified, and this is the arithmetic:

- **Peak flux.** Equation 21, at f_r because that is the worst case anywhere on the line cycle: $B_{pk} = V_{o,eff}/(4f_r N_s A_e) = 25.0 \text{ V}/(4 \times 151.75 \text{ kHz} \times 2 \times 113.1 \text{ mm}^2) = \mathbf{182 \text{ mT}}$. N_p is absent, which is the whole point of that equation.
- **Core area needed.** The same equation solved for A_e at $B_{max} = 0.20 \text{ T}$: $A_e \geq 25.0 \text{ V}/(4 \times 151.75 \text{ kHz} \times 0.20 \text{ T}) = \mathbf{103 \text{ mm}^2}$. EE6405 gives 113.1, so the margin is 1.098 — and halving N_s would double the requirement to 206 mm^2 , which EE6405 does not meet.
- **Saturation test current.** Equation 22, per unit: $I_{eq} = B_{pk} N_p A_e / L_{open} = 0.1821 \text{ T} \times 5 \times 113.1 \text{ mm}^2 / 10.33 \mu\text{H} = 9.96 \text{ A}$, specified as **14.9 A**. The tank peak is 18.3 A and handing that over instead would ask for nearly twice the flux.

The quantisation is a constraint on the tank, not a detail of the build. With 3 units in series the assembly ratio is $3N_p/N_s$, so the only ratios available are multiples of $3/N_s$. $n_T = 7.50$ exists because 15:2 exists; a tank that had asked for 7.90 could not have been wound this way at all. That has to be checked while the tank is being chosen.

Reduced to what a supplier can measure at the terminals, that becomes the sheet below. Nothing on it is a construction instruction, and L_μ is deliberately absent — Section 6.10 says why.

Item	Value	Condition
Turns ratio	$N_p : N_s = 15 : 2$	two secondary windings, centre tap outside the part
Open-circuit inductance	31.0 μ H, -4.1% at worst	primary, all other windings open
Leakage inductance	11.0 μ H $\pm 10 \%$	primary, secondary shorted
DC overlap	$\geq 90 \%$ of initial inductance	test current 15 A
Primary current	10.9 A rms / 18.3 A pk	line-cycle rms, composite peak
Secondary current, each winding	28.3 A rms / 99 A pk	line-cycle rms
Core area	$A_e \geq 103 \text{ mm}^2$	holds B_{pk} at or below 0.20 T
Switching frequency	127 to 249 kHz	at full load

Table 14: Transformer specification for the worked design.

7.9 The output bank, as sized

Both conditions of Section 6.16, worked out on the specification above.

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Condition	Asks for	Driven by
$2f_l$ ripple $\leq 5\%$	71.25 mF	I_{out} , $f_l = 47$ Hz and the allowed Δv — this one wins
Hold-up 12 ms to 19 V	67.22 mF	the energy 7.89 J between the starting voltage and 19 V, starting at the ripple trough
Selected	75.2 mF	$470\ \mu\text{F} \times 160$, the next assembly up from 71.2 mF

Table 15: The two sizing conditions, evaluated.

- **The ripple condition**, equation 24: $C_{out} \geq P_{out} / (2\pi f_{l,min} \Delta v V_{out}^2) = 657.5\ \text{W} / (2\pi \times 47\ \text{Hz} \times 0.05 \times 625\ \text{V}^2) = \mathbf{71.25\ mF}$, with Δv as a fraction and V_{out} in volts.
- **The hold-up condition**, equation 25, starting half a ripple low: $C_{out} \geq 2P_{out} T_{hold} / ((V_{out}^{-1/2} \Delta v_{pp})^2 - V_{o,min}^2) = 2 \times 657.5\ \text{W} \times 0.012\ \text{s} / (24.41^2 - 19^2) = \mathbf{67.22\ mF}$. Starting at V_{out} instead would have asked for only 59.77 mF.

The two are only 1.060 apart. That is close, so the screening inequality is worth carrying. At $k = V_{o,min} / V_{out} = 0.760$ it reads 0.373 against 0.354: the left side is larger, so ripple wins. Loosen Δv to 10 % and the sides swap.

Quantity	Value	Against
Achieved ripple	1.18 V (4.74 %)	5 % allowed
Achieved hold-up	13.42 ms	12 ms required — and 15.10 ms if it had been started at V_{out} instead of at the trough
Ripple current, switching component	23.8 A rms	—
Ripple current, $2f_l$ component	18.6 A rms	orthogonal to the above
Ripple current, total	30.24 A rms	0.19 A in each of 160 parts
Bank ESR	0.1875 mΩ	30 mΩ each, 160 in parallel
Ripple and noise	17.4 mV	with the 10 μF ceramic bypass

Table 16: What the selected bank then delivers, and what it has to survive.

7.10 Where the power goes

The specification allowed 27.3 W of loss to get from $P_{in} = 684.8\ \text{W}$ to $P_{out} = 657.5\ \text{W}$. That allowance was split three ways before any component existed, and the itemised device losses are computed afterwards. The two do not have to agree, and here they do not.

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Item	Value	Note
Budget — input bridge	5.52 W	synchronous bridge assumed
Budget — EMI filter	8.34 W	—
Budget — LLC stage	13.42 W	$\eta_{HB} = 98\%$ of $P_{in,LLC}$
Primary conduction, standing device	9.65 W	the half-bridge leg that never switches — the single largest item
Primary switching, three positions	4.83 W	not a problem
Secondary rectifiers	5.65 W	1.411 W per device, 2 in parallel per leg
Current-sense resistors	2.86 W	4.54 W at the worst switching cycle
Output capacitor ESR	0.162 W	—
Itemised total	23.15 W	and the magnetics are not in it

Table 17: The budget as assumed, and the device losses as computed. The second block is not a breakdown of the first.

That disagreement is the efficiency assumption being caught. The itemised device losses already come to 23.2 W against the 13.4 W that $\eta_{HB} = 98\%$ allows the LLC stage, and the transformer is not even in the list. So the assumption is optimistic, which is why it is filed as an assumption rather than as a figure. It is not load-bearing: taking 95 % instead moves R_{ac} and R_{CS} by about 3 % and nothing downstream is sensitive to that. What it does change is the thermal design, and that is settled by measurement rather than by either number.

7.11 What the semiconductors have to be

Requirements, not part numbers. What to pick depends on package, thermal design and cost. The four rules of Section 6.17 decide whether these are even stated correctly.

Item	Requirement
Primary drain-source voltage	≥ 485 V, so a 600 V class part
Primary body diode	fast recovery: in full bridge the body diode conducts before turn-on
Primary $R_{DS(on)}$ per device	≤ 25.2 m Ω hot to meet the loss budget
Primary $C_{o(tr)}$	small enough that the tank can swing the node inside 220 ns
Secondary drain-source voltage	≥ 80 V including the centre-tap doubling
Secondary rectifier current	28.34 A rms per leg, 98.6 A peak
Secondary package	check the lead and clip rating, not only the die

Table 18: Semiconductor requirements for the worked design.

7.12 The voltage loop, as built

Section 6.19 gives the method. These are the numbers it produces on this converter, in the order they are found.

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#	Quantity	Value	From
1	Output divider	R_1 100 k Ω / R_o 11 k Ω	sets the regulated output against the 2.495 V reference
2	Plant gain G_o	130.06 rad/s	a pure integrator — the output is a capacitor fed by a power source
3	Error-amplifier gain needed at $2f_l$	0.4541	what the loop needs there, against what the amplifier has
4	K factor K_v	2.7218	places the zero and pole symmetrically
5	Geometric mean f_{MB}	16.921 Hz	the point they are placed about
6	Zero / pole	6.217 / 46.057 Hz	f_{MB}/K_v and $K_v f_{MB}$
7	High-frequency pole f_{px}	999.4 Hz	added to roll off switching noise
8	Components	C_{Fo} 82 nF, C_F 560 nF, R_F 47 k Ω , C_{fx} 1.55 nF	the nearest standard values to the pole and zero above
9	Optocoupler bias	R_P 1.3 k Ω , R_B 6.2 k Ω	R_B must land inside 5.76 to 6.48 k Ω or the anti-saturation circuit cannot act

Table 19: The compensator, from the plant to the parts.

Quantity	Value	Against
Crossover f_{cross}	17.57 Hz	the high teens of hertz this topology lands in
Phase margin	49.64°	45° as a floor
Gain margin	34.18 dB at 202 Hz	6 dB floor, 10 dB comfortable
Third harmonic on the input current	4.23 %	the price of crossing this low
Feedback ripple at the burst point	52 mV	move R_{BM} down 2.60 k Ω to clear it

Table 20: And what that loop then measures as.

7.13 The parts around the controller

These are the results. The sections after this one derive each value in the order it has to be done, because every one of them is sized against something already fixed.

Part	Value	Result
C_T / R_T	470 pF / 11 k Ω	f_{Min} 92.2 kHz, f_{Max} 319.6 kHz
R_{CS}	24.0 m Ω (120 m Ω $\times$ 5 in parallel)	OCP1 at 1.251 $\times$ the composite peak
R_{CFG}	30 k Ω	V_{BO} 84.8 V, morphing enabled
R_{BM}	30 k Ω	burst-mode entry point
ZCD divider	330 k Ω / 19 k Ω	OVP1 28.16 V, OVP2 30.61 V
C_{in}	2200 nF film	about 3.2 nF/W — there is no bulk capacitor
Compensation	82 nF / 560 nF / 47 k Ω / 1.55 nF	f_{cross} 17.57 Hz, ϕ_M 49.64°

Table 21: Controller network for the worked design.

7.14 The oscillator: C_T first, then R_T

The VCO charges C_T from the sum of a fixed current V_{ref}/R_T and the error-amplifier current I_{EA} , up to $V_{ref} = 1.5$ V, then adds a fixed idle time:

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$$\frac{T_{sw}}{2} = \frac{C_T V_{ref}}{\frac{V_{ref}}{R_T} + I_{EA}} + T_{idle} \quad (33)$$

I_{EA} is the only actuator in the converter. More current means a higher frequency, which means less power. Everything the control loop does, it does through this one term.

C_T is chosen first because it sets the *span*, and R_T after it because it sets the *floor*:

$$C_{T,max} = \frac{I_{EA,max}}{2V_{ref}} \cdot \frac{(1 - 2T_{idle}f_{sw,max})(1 - 2T_{idle}f_{sw,min})}{f_{sw,max} - f_{sw,min}}, \quad C_{T,min} = \frac{1}{R_{T,max}} \left(\frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (34)$$

$$R_{T,ceil} = \frac{1}{C_T} \left(\frac{1}{2f_{sw,min}} - T_{idle} \right) \quad (35)$$

with $f_{sw,min}$ set to f_o — the frequency floor of the whole design, for the reason in Section 4.2. Here that window is 176 to 703 pF, the datasheet allows 270 to 1000 pF, and the part is taken from the middle of the overlap rather than from either edge: $C_T = 470$ pF, COG. Then $R_{T,ceil} = 11.24$ kΩ gives $R_T = 11$ kΩ.

$R_{T,ceil}$ is a maximum, not a minimum, whatever the name suggests. $f_{Min} = 1/[2(C_T R_T + T_{idle})]$ falls as R_T grows. Round this number up and the VCO clamp drops below f_o , leaving only the anti-capacitive protection between the converter and the capacitive region. Round it **down**.

The three frequencies the selected pair actually produce are

$$f_{Min} = \frac{1}{2(C_T R_T + T_{idle})}, \quad f_{Max} = \frac{1}{2\left(\frac{C_T}{\frac{I_{EA,max}}{V_{ref}} + \frac{1}{R_T}} + T_{idle}\right)} \quad (36)$$

Putting the selected pair into the first of those, with $T_{idle} = 250$ ns: $f_{Min} = 1/[2(470 \text{ pF} \times 11 \text{ kΩ} + 0.25 \text{ μs})] = 1/(2 \times 5.420 \text{ μs}) = 92.25$ kHz, which is what has to clear f_o .

Quantity	Value	Limit	Margin
f_{Min}	92.2 kHz	above $f_o = 90.4$ kHz	1.021
f_{Max}	319.6 kHz	above $f_{sw,max} = 249.0$ kHz	1.283
f_{SU}	503.5 kHz	below the 675 kHz silicon ceiling	1.341
$R_T C_T$	5.17 μs	2.5 to 12 μs	2.068 / 2.321
R_T	11 kΩ	5 to 30 kΩ	2.200 / 2.727
C_T	470 pF	270 to 1000 pF	1.741 / 2.128

Table 22: Oscillator: what the two parts produce, and the limits each result has to clear.

T_{idle} is the weakest number in this chapter. The draft datasheet gives 700 ns in the text and a table that back-solves to 250 ns; this design is worked at 250 ns. At 700 ns the same R_T and C_T put f_{Min} essentially on top of f_o and the floor margin disappears. **It is not a precision question, it is whether the two clamps still bracket the operating range**, and it is the first thing to measure on hardware.

7.15 The current-sense resistor, which sets three things at once

R_{CS} is asked for by two conditions and the smaller one wins:

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$$R_{CS, max1} = \frac{16.8 \Omega \cdot W}{P_{in}} \quad R_{CS, max2} = \frac{0.55 V}{I_{Lr, pk}} \quad (37)$$

The first is the maximum-power law of Section 4.4; the second is the OCP1 trip point. Opened out, equation 37 is two divisions — $16.8/P_{in}$ and $0.55/I_{Lr, pk}$ — and they give 24.53 and 30.03 mΩ, so the power law binds. Dissipation then fixes how many resistors that has to be split over. At the worst switching cycle that is 4.54 W, so **5 in parallel**; 120 mΩ each gives **$R_{CS} = 24.0 \text{ m}\Omega$** .

The denominator of $R_{CS, max2}$ is the composite tank peak, not the reflected load current. R_{CS} sits in the bridge return, so it carries the magnetising component too. Using $I_{trafo, pk} = 16.37 \text{ A}$ instead of $I_{Lr, pk} = 18.32 \text{ A}$ loosens the limit by about 12 % and the over-current protection with it.

Result	Value	Against
Maximum input power	700.0 W	$P_{in} = 684.8 \text{ W}$
OCP1 trip	22.92 A	composite peak 18.32 A, margin 1.251
OCP2 trip	31.25 A	immediate stop, 50 μs, restart at f_{SU}
Sense dissipation	4.54 W total, 0.909 W each	1 W parts, margin 1.100

Table 23: What the selected R_{CS} then fixes.

7.16 Burst mode: one resistor, read once at power-up

$$R_{BM} = 16.7 \frac{k\Omega}{V^2} R_{CS} P_{in, BM}, \quad V_{BM, eq} = \frac{R_{BM}}{100 k\Omega} + 0.5 V \quad (38)$$

Entering burst at 75 W of input power — about 11 % of rated — asks for 30.06 kΩ, so **$R_{BM} = 30 \text{ k}\Omega$** and the equivalent threshold on the FB pin is 0.800 V. The valid range is 15 to 140 kΩ. Tie the pin to ground and burst mode is off altogether; deep burst stays active either way.

This threshold and the feedback ripple have to be checked against each other, or the converter chatters in and out of burst once per half line cycle — Section 6.21.

7.17 Brown-out and bridge configuration: the CFG pin

One resistor does two unrelated jobs, and both are read once at power-up:

$$V_{BO, pk} = R_{CFG} \times 4 \frac{V}{k\Omega}, \quad V_{BO, rms} = \frac{V_{BO, pk}}{\sqrt{2}} \quad (39)$$

The brown-out threshold is compared against the *peak* of the mains, so it has to be converted before it is compared with an rms specification — a $\sqrt{2}$ that is easy to drop. Asking for brown-out just below the minimum line gives $R_{CFG, max} = 31.82 \text{ k}\Omega$; **this is a maximum, so it rounds down, to 30 kΩ**, and $V_{BO} = 84.85 \text{ V}_{rms}$ against a 90 Vac minimum — margin 1.061. Rounding up instead would stop the converter starting at low line.

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R_{CFG}	LOUT2	Configuration
15 k Ω	open	morphing, fixed brown-out (60 V / 70 V peak)
15 to 47 k Ω	open	morphing, adjustable brown-out — this design
47 to 100 k Ω	open	fixed full bridge
15 k Ω	to GND	fixed half bridge, split C_r , fixed brown-out
15 to 100 k Ω	to GND	fixed half bridge, split C_r , adjustable brown-out

Table 24: What R_{CFG} and LOUT2 select together. The 235 and 245 V thresholds are fixed inside the IC and cannot be moved.

For a universal-input design there is only one window. The datasheet floor is 15 k Ω and brown-out at the minimum line is the ceiling at 31.8 k Ω , so the 47 k Ω morphing limit never comes into play. And hang nothing on LOUT2. The pin-strap drives 300 μ A into it and needs 2.5 V, so any pull-down under about 8 k Ω reads as a request for a fixed half bridge.

7.18 Output sensing and over-voltage: the ZCD divider

The auxiliary winding is divided down to the ZCD pin, and the divider ratio alone sets both over-voltage thresholds — the absolute resistor values only set the bias current:

$$R_{ZCD,L} = \frac{2.3 \text{ V}}{I_{bias}}, \quad R_{ZCD,H} = R_{ZCD,L} \left(\frac{n_{aux}}{n_{sec}} \frac{V_{OVP1,out}}{2.3 \text{ V}} - 1 \right) \quad (40)$$

$$V_{OVP1} = \frac{2.3 \text{ V}}{n_{aux}/n_{sec}} \left(\frac{R_{ZCD,H}}{R_{ZCD,L}} + 1 \right), \quad V_{OVP2} = \frac{2.5}{2.3} V_{OVP1} \quad (41)$$

Aiming OVP1 10 % above the output gives 27.50 V as a target. With $n_{aux}/n_{sec} = 1.5$ the calculated pair is 19.17 / 321.8 k Ω . The parts fitted are **19 k Ω and 330 k Ω** . The upper one is deliberately the *next E24 value above* the calculation, so OVP1 lands clear of the ripple crest rather than on it. The result is OVP1 28.16 V and OVP2 30.61 V, and start-up hands over to the loop at 16.53 V of output.

Two ways to get this wrong, both of which stop the converter dead. Swapping the two resistors inverts the ratio and OVP1 trips at well under a volt of output, so the converter never starts. And the winding polarity must make ZCD *positive* when the low side of leg 1 is on; reversed, the bridge hard switches from the first pulse.

The turns-ratio ceiling in the datasheet does not apply here. It asks that the rectified auxiliary voltage stay under the V_{CC} rating at OVP2, which is a question only when that winding supplies V_{CC} . In this design V_{CC} comes from a housekeeping rail and the winding senses only, so its voltage is free — 37.5 V nominal, 45.9 V at OVP2 — and the only remaining constraint is that the turns come out whole. Carrying that check over from a self-supplied design reports a failure that is not one.

7.19 HVSU and V_{CC}

- **HVSU connects ahead of the input bridge**, on the ac side, through one 1000 V diode per line. Behind the bridge both the X-capacitor discharge and the brown-out detection stop working, because neither can see the mains disappear.
- The V_{CC} capacitor has to hold the IC from $V_{CCon} = 17 \text{ V}$ down to $V_{CCoff} = 8 \text{ V}$ for as long as it takes the supply to take over. The start-up unit stops 120 ms after V_{CCon} is reached, whether or not anything has.

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- Where a linear regulator feeds V_{CC} , put a diode between its output and the pin so the start-up charging current cannot be pushed back into it.
- A 100 nF bypass sits at the pin itself.

7.20 Controller pin rules

These follow from the datasheet and each one causes immediate misbehaviour if broken.

Pin	Rule
ISEN	No filter and no series resistor. The maximum-power law and the over-current threshold share this pin.
CFG, BM	No capacitor. Both are strapped at power-up. An out-of-range CFG resistor latches the controller off.
DRV_EN, RT	No capacitor. Both are sampled briefly every few milliseconds and a capacitor is read as a fault.
LOUT2	No pull-down if morphing is used. A few kilohms to ground reads as a fixed half bridge.
HVSU	Connect ahead of the input bridge, on the ac side. Behind the bridge, both X-capacitor discharge and brown-out detection stop working.
ZCD	Auxiliary winding polarity must make ZCD positive when the low side of leg 1 is on. Reversed, the bridge hard switches.
HOUTx, LOUTx	Logic-level outputs, not gate drivers. External half-bridge drivers are required.

Table 25: Pin rules that must not be broken.

7.21 Protections, and where each threshold is set

Every one of these is a design constant fixed by a resistor this note has already chosen, so a change made for one reason moves a threshold set for another. The table is here because the thresholds are otherwise spread over five sections.

Protection	Set by	This design
Brown-out, on the rectified mains	R_{CFG} , read at power-up	V_{BO} 84.8 V, clear of the 90 Vac minimum by 1.061
Cycle-by-cycle over-current, OCP1	R_{CS} — the same resistor as the maximum-power law	22.92 A against a composite peak of 18.32 A, a margin of 1.251
Second-level over-current, OCP2	R_{CS} , fixed ratio to OCP1	31.25 A
Output over-voltage, OVP1	the ZCD divider on the auxiliary winding	28.16 V, above the ripple crest and below OVP2
Second-level over-voltage, OVP2	the same divider	30.61 V
Capacitive-mode protection	internal; it watches the ZCD edge against the gate	the last line of defence only — the frequency floor f_{Min} is the first
Burst mode at light load	R_{BM} , read at power-up	30 k Ω ; the $2f_l$ feedback ripple is 52 mV, so see Section 6.21
Loss of ZVS at the worst corner	not a protection — a design margin	swept, not estimated; Section 6.5

Table 26: Protections and the component that sets each one.

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Two of these share one resistor. R_{CS} sets the over-current thresholds *and* the maximum-power law, because the controller reads both from the same pin. Changing it to fix a power limit moves OCP1, and the other way round. That is also why the pin tolerates no filter.

7.22 Ten things that go wrong

- Designing the tank for the mains range instead of the **equivalent** range. The corners are the morphing edges.
- Carrying over λ from a two-stage LLC. L_m/L_r of 5 to 10 will not start at low line.
- Putting the oscillator floor below f_o and relying on the anti-capacitive protection to catch it.
- Checking the turns ratio instead of the reflected voltage, so that n and n_T are with no warning interchanged.
- Rating the primary switch on the reflected load current rather than the composite tank peak.
- Omitting the $2f_i$ component from the output capacitor ripple current, or judging it per winding on a centre tap.
- Rounding L_m up, which without showing it removes the ZVS margin.
- Trusting the closed-form ZVS estimate instead of sweeping the selected tank — or sweeping only the design corner, which is where the *gain* is worst and not always where ZVS is.
- Using capacitor ESR at 120 Hz, or $R_{DS(on)}$ at 25 °C, to compute a loss at the operating point.
- Judging device stress without dividing by the number of parts actually in parallel.

7.23 System design rules

The same material as a checklist, in the order a design meets it. Every line is argued for somewhere above; collected here so a reviewer can run down the list without re-reading the argument.

- **Design the tank at the real ends of the equivalent range.** With morphing that is 173.2 and 332.3 Vac, not 2×90 and 264. Looking only at the mains range misses both worst corners.
- **Expect λ near 0.5.** The 5-to-10 inductance ratio of a two-stage LLC will not start at low line.
- **Keep the oscillator floor above f_o ,** and treat the anti-capacitive protection as the last line, not the first.
- **Size the output bank from both conditions** — ripple and hold-up — take the larger, and then check the rms ripple current separately. The capacitance and the current are different questions.
- **Put the voltage-loop crossover in the tens of hertz,** then deal with the feedback ripple that survives by moving R_{BM} , not by speeding the loop up.
- **Measure both morphing edges,** at full load and at light load. The half-bridge edge is the worst case for gain and ZVS, the full-bridge edge for switching frequency.
- **Check the turns ratio as a reflected voltage,** $nV_{o,eff}$, never as a bare ratio — that is what keeps n and n_T from being interchanged.
- **Judge the over-current margin on the composite tank peak,** not on the reflected load current.
- **Include the $2f_i$ component** in the output capacitor ripple current, and judge a centre tap at the output node rather than per winding.
- **Carry the conduction ratio $\sqrt{d}$ into the secondary rms.** Dropping it overstates the rms by about 12 % here, and the loss — which goes as the square — by about 26 %.
- **Decide how L_r is built before specifying the transformer,** and put open-circuit and short-circuit inductance on the drawing alongside the turns.
- **Use capacitor ESR at the switching frequency,** not the 120 Hz $\tan\delta$.
- **Never round L_m up.** It lowers λ and takes the ZVS margin with it.
- **Verify ZVS by sweeping the selected tank,** not with the closed form — and if the closed form is used anyway, feed it λ_{act} and the Q of the moment, and know that its error has no fixed sign.

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- **Record calculated and selected values separately**, and make every downstream check read the selected one. A sheet that with no warning reads the calculated value reports margins the hardware does not have.

7.24 What to measure first on hardware

In order. The first four decide whether the design is sound at all; the rest confirm margins that are computed but never yet seen.

- **$f_{sw}(\theta)$ over a line half cycle.** This settles T_{idle} , and with it both oscillator clamps — the thinnest margin in the design. Everything else in the controller network hangs off it.
- **Cold start into the full output bank.** A 75.2 mF bank is a far heavier start-up load than a conventional design presents, and the start-up window is finite. This risk is created by the architecture, so it has no precedent to borrow from.
- **ZVS at the half-bridge morphing edge** (245 V_{pk}, full load), on the gate and mid-point waveforms. The calculation says 460 ns against 220 ns; confirm it is not hard switching, and that the adaptive dead time actually settles where it is assumed to. **Then measure the worst point of the sweep as well** — 25 % load at the 332.3 Vac equivalent corner, 401 ns — because that, not the design corner, is the figure this design is held to.
- **Switching frequency at the full-bridge morphing edge** (235 V_{pk}, full load). Calculated 249.0 kHz; check it does not run into the VCO ceiling and get power-limited.
- **Input current and THD at 90 Vac, full load**, to put a number on the zero-crossing dead zone.
- **Output ripple against what the load actually tolerates.** The bank is sized from a specification; this is where the specification is tested.
- **Ripple current and temperature rise in the output bank** — 30.24 A rms calculated, 0.19 A per capacitor.
- **Device temperatures, primary and secondary.** This is where the accepted loss-budget miss is settled: measure it in **half-bridge morphing**, above 245 V_{pk}, because in full bridge the standing device is switching and dissipates far less.
- **Feedback ripple against the burst threshold** at light load.
- **An ac dropout at the worst line phase** — cut the mains at the trough of the ripple and confirm the output is still above 19 V after 12 ms.
- **A step across the morphing band**, both directions, on a programmable source. A ramp gives the loop time to follow and hides the transition.

8 References

- STMicroelectronics, *L6790A LLC-PFC controller*, preliminary datasheet, 30 April 2026. **Draft.**
- STMicroelectronics, *EVL6790_670W* evaluation board schematics.
- Infineon Technologies, *LLC Converter Design Note*, AN 2013-03, V1.0, March 2013.
- Infineon Technologies, *Resonant LLC converter: operation and design*, AN 2012-09, V1.0, September 2012.
- ON Semiconductor / Fairchild, *Half-bridge LLC resonant converter design using FSFR-series Fairchild power switch*, AN-4151.
- Monolithic Power Systems, *Understanding LLC operation*, part 2.
- Toshiba Electronic Devices & Storage, *Power factor correction (PFC) circuits*, application note.
- ROHM Semiconductor, TechWeb, *LLC operating regions* (region framing of Figure 9).
- W. Wenbo et al., *A single-stage 1.65 kW ac-dc LLC converter*, IEEE ECCE.

9 Errata and open items

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Everything in this chapter is something the note works around rather than something it solves. It is last because none of it is needed to follow the design, and first to be re-read before a production release — most of it is a question that only the released datasheet or a prototype can close.

9.1 Datasheet: the draft is not self-consistent

Item	What the draft says	What is used here, and why
Oscillator idle time T_{idle}	Section 5.3.2 says 700 ns; Table 5 back-solves to about 250 ns from the frequency expressions	250 ns. The two clamps bracket the operating range at 250 ns and stop doing so at 700 ns, so this is a measurement, not a rounding
Brown-out expression	$V_{BO} = \min(60 \text{ V}, R_{CFG} \cdot 4 \text{ V/k}\Omega)$ — read literally this is 60 V for every resistor, which contradicts the configuration table	Read as $R_{CFG} \cdot 4 \text{ V/k}\Omega$ with 15 k Ω as the floor. That is the only reading consistent with the table
Brown-out units	the threshold is a <i>peak</i> voltage; board notes and design tools quote rms	Converted explicitly wherever it is compared ($\sqrt{2} = 1.414$)
Thermal resistance	TBD	No junction temperature can be predicted from the datasheet alone; the loss budget is checked against measured rise instead
ZCD absolute maximum	lower limit given as TBD	The divider is sized from the OVP thresholds only
Driver naming in Section 5.3.1	LOUT1 and LOUT2 are transposed in one sentence against the block diagram and the configuration table	The block diagram and the table are taken as correct: LOUT2 is the pin that is strapped and held high in half bridge

Table 27: Points where the draft datasheet contradicts itself or leaves a value open. Every one of them must be re-checked against the released document.

9.2 Traps in the obvious way of building a design sheet

These are not datasheet problems. They are places where the obvious way to set a design sheet up gives an answer that is wrong in a way nothing flags, so they are worth stating as traps rather than as corrections to any particular tool.

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Where	The trap	Consequence if missed
Equivalent input range	taking the maximum from the ac maximum instead of the full-bridge morphing edge	The frequency corner is never evaluated. Here it is the difference between 264 and 332.3 Vac equivalent, and it also sets λ
λ from the minimum-gain condition	the same corner again, one step earlier	The required λ comes out far too small — a factor of over a hundred in one case — and L_m with it
Line-cycle rms currents	evaluating one phase, usually $\theta = \pi/4$, and calling it the line-cycle value	The output bank ripple current and the rectifier loss are both understated. A Simpson rule over the half cycle costs nothing and is exact enough
Secondary rms above resonance	squaring a full half sine when the current is truncated	Up to 8 % at the high corner. Below resonance the factor is exactly 1, which is why it can sit unnoticed for a long time
Primary device rating	rating on the reflected load current $I_{\text{trafo,pk}}$	About 12 % under-rated — the switch carries the <i>composite</i> tank current
Loss and thermal resistance	computing loss from the 25 °C $R_{\text{DS(on)}}$ and then asking for a heatsink that holds 125 °C	The required thermal resistance comes out roughly twice as easy as it is
ZVS check	the closed-form shortcut, at the design Q and with the design λ	Error of either sign; on this tank 36 % conservative, on another about 23 % optimistic. Optimistic is the dangerous direction
Selected versus calculated	a check that reads the calculated value while the board carries the selected one	Every downstream margin is reported for a design that was not built

Table 28: Traps that produce a plausible wrong number.

9.3 Constants used here without a derivation

- The **$\pi^2/8$** in the dead-time form of the λ condition. It behaves like a correction for the fundamental content of a square wave, but no source states it. It is indicative; the binding check is the ZVS sweep.
- The **exponent 5** in the closed-form ZVS shortcut. This is a fitted number, and it is the main reason the shortcut errs unpredictably near the boundary.
- The **0.744** in the input-voltage margin factor of the compensator design. It scales the crossover target and nothing downstream is sensitive to it at the percent level.
- The **16.8 $\Omega \cdot W$** maximum-power constant is consistent with the feedback span and the multiplier gains quoted in the datasheet — $2.8 \text{ V} / 0.167 = 16.8$ — so it is derived rather than assumed, and is listed here only because the gains themselves are draft values.

9.4 Open items in this design

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Item	State	What closes it
T_{idle}	two candidate values, 250 and 700 ns	Measure $f_{sw}(\theta)$ on the first board
Primary conduction loss	0.311 against the budget — not met, and accepted	No single 600 V device meets a 3 W budget in the standing position. Whether it needs a heatsink is a thermal measurement, not a calculation
Secondary loss budget	1.063 — essentially exhausted	One more device in parallel per leg recovers it; the decision waits on the measured temperature
Transformer inductance tolerance	the tank tolerates a 4.1 % fall in open-circuit inductance and the usual specification asks for ± 10 %	Either tighten the low side with the supplier, or lower R_T to lift f_{Min} first. Before the transformer is ordered
Output bank volume and height	75.2 mF, 160 parts	A mechanical question, not an electrical one, and it can send the whole architecture back to the output voltage
Standby power	burst entry assumed at 75 W	Measure, then trade R_{BM} against the feedback ripple
Efficiency assumption	$\eta_{HB} = 98$ % assumed	Optimistic. Taking 95 % instead moves R_{CS} and R_{ac} by about 3 %, so nothing downstream is sensitive — but it should be replaced by a measurement

Table 29: What is not settled, and what would settle it.

Every cross-check here is a consistency check, not a correctness check. The numbers agree across three independent implementations, which means the three implement the same equations — not that the equations describe the hardware. That is what the measurement list in Section 7.24 is for.